

Electronic Devices and Circuits

Previous Year Questions • 1987–2026

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Electronic Devices and Circuits (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience



Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry



> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

Important Links & Resources

Visit our Website

prepfusion.in

Solutions, practice & report

pyq.prepfusion.in

Subscribe: English Channel

youtube.com/@PrepFusion_GATE

Subscribe: Hindi Channel

youtube.com/@prepfusion-hindi

Join our Telegram group

t.me/All_About_Learning

Follow us on LinkedIn

linkedin.com/company/prepfusionee

Subscribe: Himanshu Agarwal

youtube.com/@HimanshuAgarwal_

Subscribe: Anish Saha

youtube.com/@AnishSaha_

Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A, Suresh Ravanam, Aabhas Anuraag Suman and Piyush Raj.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

Contents

Sl.	Topic	Pages
1	Basic Semiconductor Physics	1
2	PN Junctions	14
3	BJTs	24
4	MOSCap and MOSFETs	29
5	Optoelectronic Devices	37

PrepFusion

UNIT

1

Basic Semiconductor Physics

Topics

1. Physics and Fundamentals
2. Carrier Transport Phenomena
3. Semiconductor Under Non-Equilibrium
4. Fermi Dirac Distribution and Energy Band Diagrams

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 74	**MCQ:** 54	**MSQ:** 4	**NAT:** 16
1M: 46	**2M:** 28	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2020)–6M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 2M

Consider that the concentration of electrons in a semiconductor bar varies linearly from 2×10^{17} cm^{-3} at $x = 1$ μm to 1×10^{16} cm^{-3} at $x = 4$ μm along the x-direction. Assume that the concentration of electrons is not varying along other directions (that is along y- and z-directions). [Given: the mobility of electron is 1400 cm^{2}V^{-1}s^{-1}, thermal voltage is 25 mV and electronic charge is 1.6×10^{-19} C.] The density of electron diffusion current (in A/mm^2) is ___. (rounded off to two decimal places)

Q1.25.1 MSQ 2025 1M

Which of the following can be used as an n-type dopant for silicon? Select the correct option(s).

- ☐ **A.** Arsenic
- ☐ **B.** Boron
- ☐ **C.** Gallium
- ☐ **D.** Phosphorous

Q1.25.2 MCQ 2025 2M

The intrinsic carrier concentration of a semiconductor is 2.5×10^{16} /m^3 at 300 K. If the electron and hole mobilities are 0.15 m^2/Vs and 0.05 m^2/Vs, respectively, then the intrinsic resistivity of the semiconductor (in kΩ.m) at 300 K is _____.
(Charge of an electron $e = 1.6 \times 10^{-19}$ C.)

A. 1.65 **B.** 1.25 **C.** 0.85 **D.** 1.95

Q1.25.3 MCQ 2025 2M

The electron mobility μ_n in a non-degenerate germanium semiconductor at 300 K is 0.38 m^2/Vs. The electron diffusivity D_n at 300 K (in cm^2/s, rounded off to the nearest integer) is _____.
(Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K and the charge of an electron $e = 1.6 \times 10^{-19}$ C.)

A. 26 **B.** 98 **C.** 38 **D.** 10

Q1.24.1 MSQ 2024 1M

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the points A, B, and C mark three different positions. Which of the following statements is/are true?

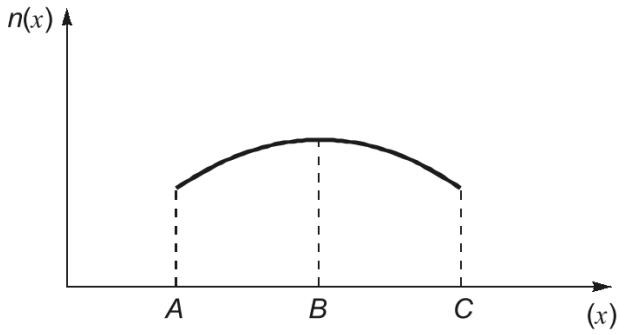


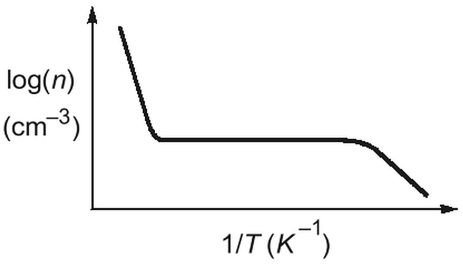
Fig. Q1.24.1

- ☐ **A.** For x between B and C, the electron diffusion current is directed from C to B.
- ☐ **B.** For x between B and A. the electric field is directed from A to B.
- ☐ **C.** For x between B and A, the electron drift current is directed from B to A.
- ☐ **D.** For x between B and C, the electric field is directed from B to C .

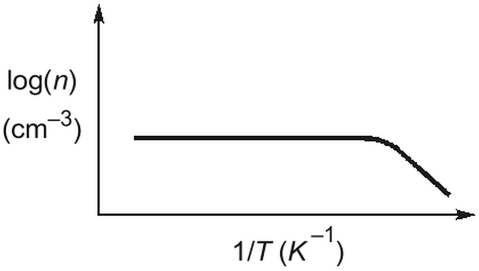
Q1.24.2 MCQ 2024 1M

For non-degenerately doped n-type silicon, which one of the following plots represent the temperature (T) dependence of free electron concentration (n)?

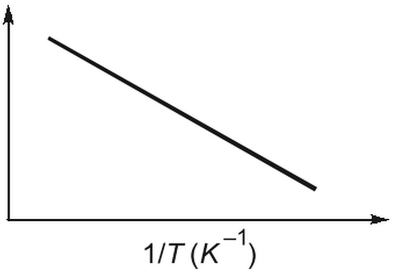
A.



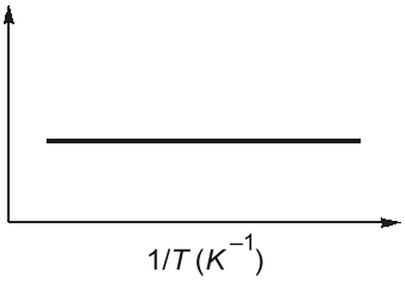
B.



C.



D.



Q1.24.3 NAT 2024 2M

A non-degenerate n-type semiconductor has 5% neutral dopant atoms. Its Fermi level is located at 0.25 eV below the conduction band (E_C) and the donor energy level (E_D) has a degeneracy of 2. Assuming the thermal voltage to be 20 mV. The difference between E_C and E_D (in eV, rounded off to two decimal places) is _____.

Q1.23.1 MCQ 2023 1M

In a semiconductor, if the Fermi energy level lies in the conduction band, then the semiconductor is known as

A. degenerate n-type. **B.** degenerate p-type.
C. non-degenerate n-type. **D.** non-degenerate p-type.

Q1.23.2 MCQ 2023 1M

For an intrinsic semiconductor at temperature $T = 0$ K, which of the following statement is true?

A. All energy states in the valence band are filled with electrons and all energy states in the conduction band are empty of electrons.

B. All energy states in the valence band are empty of electrons and all energy states in the conduction band are filled with electrons.

C. All energy states in the valence and conduction band are filled with holes.

D. All energy states in the valence and conduction band are filled with electrons.

Q1.23.3 NAT 2023 2M

In an extrinsic semiconductor, the hole concentration is given to be $1.5n_i$ where n_i is the intrinsic carrier concentration of 1×10^{10} cm^{-3}. The ratio of electron to hole mobility for equal hole and electron drift current is given as _____ (*rounded off to two decimal places*).

Q1.23.4 NAT 2023 2M

In a semiconductor device, the Fermi-energy level is 0.35 eV above the valence band energy. The effective density of states in the valence band at $T = 300$ K is 1×10^{19} cm^{-3}. The thermal equilibrium hole concentration in silicon at 400 K is _____ $\times 10^{13}$ cm^{-3} (*rounded off to two decimal places*).
Given kT at 300 K is 0.026 eV.

Q1.22.1 MSQ 2022 2M

Select the correct statement(s) regarding semiconductor devices:

☐ **A.** Electrons and holes are of equal density in an intrinsic semiconductor at equilibrium.

☐ **B.** Collector region is generally more heavily doped than base region in a *BJT*.

☐ **C.** Total current is spatially constant in a two terminal electronic device in dark under steady-state condition.

☐ **D.** Mobility of electrons always increases with temperature in silicon beyond 300 K.

Q1.22.2 MCQ 2022 1M

Consider a long rectangular bar of direct bandgap p-type semiconductor. The equilibrium hole density is 10^{17} cm^{-3} and the intrinsic carrier concentration is 10^{10} cm^{-3}. Electron and hole diffusion lengths are 2 μm and 1 μm, respectively.
The left side of the bar ($x = 0$) is uniformly illuminated with a laser having photon energy greater than the bandgap of the semiconductor. Excess electron-hole pairs are generated ONLY at $x = 0$ because of the laser. The steady state electron density at $x = 0$ is 10^{14} cm^{-3} due to laser illumination. Under these conditions and ignoring electric field, the closest approximation (among the given options) of the steady state electron density at $x = 2$ μm, is _____.

A. 0.37×10^{14} cm^{-3} **B.** 0.63×10^{13} cm^{-3}
C. 3.7×10^{14} cm^{-3} **D.** 10^{3} cm^{-3}

Q1.22.3 MCQ 2022 1M

In a non-degenerate bulk semiconductor with electron density $n = 10^{16}\ cm^{-3}$, the value of $E_C - E_{F_n} = 200\ meV$, where E_C and E_{F_n} denote the bottom of the conduction band energy and electron Fermi level energy, respectively. Assume thermal voltage as 26 meV and the intrinsic carrier concentration is $10^{10}\ cm^{-3}$. For $n = 0.5 \times 10^{16}\ cm^{-3}$, the closest approximation of the value of $(E_C - E_{F_n})$, among the given options, is ____

A. 226 meV **B.** 174 meV
C. 218 meV **D.** 182 meV

Q1.22.4 NAT 2022 2M

A p-type semiconductor with zero electric field is under illumination (low level injection) in steady-state condition. Excess minority carrier density is zero at $x = \pm 2l_n$, where $l_n = 10^{-4}$ cm is the diffusion length of electrons. Assume electronic charge, $q = -1.6 \times 10^{-19}$ C. The profiles of photo-generation rate of carriers and the recombination rate of excess minority carriers (R) are shown. Under these conditions, the magnitude of the current density due to the photo-generated electrons at $x = \pm 2l_n$ is ___ mA/cm^2 (Rounded off to two decimal places).

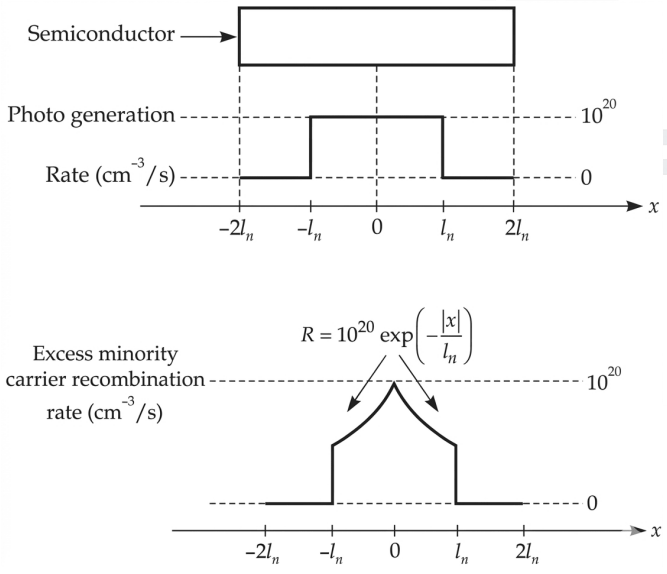


Fig. Q1.22.4

Q1.21.1 MCQ 2021 1M

A bar of silicon is doped with boron concentration of 10^{16} cm^{-3} and assumed to be fully ionized. It is exposed to light such that electron hole pairs are generated throughout the volume of the bar at the rate of 10^{20} cm^{-3} s^{-1}. If the recombination lifetime is 100 μs, intrinsic carrier concentration of silicon is 10^{10} cm^{-3} and assuming 100% ionization of boron, then the approximate product of steady-state electron and hole concentrations due to this light exposure is,

A. 10^{32} cm^{-6} **B.** 2×10^{20} cm^{-6}
C. 2×10^{32} cm^{-6} **D.** 10^{20} cm^{-6}

Q1.21.2 MCQ 2021 1M

The energy band diagram of a p-type semiconductor bar of length L under equilibrium condition (i.e.. the Fermi energy level E_F is constant) is shown in the figure. The valance band E_V is sloped since doping is non-uniform along the bar. The difference between the energy levels of the valence band at the two edges of the bar is Δ.

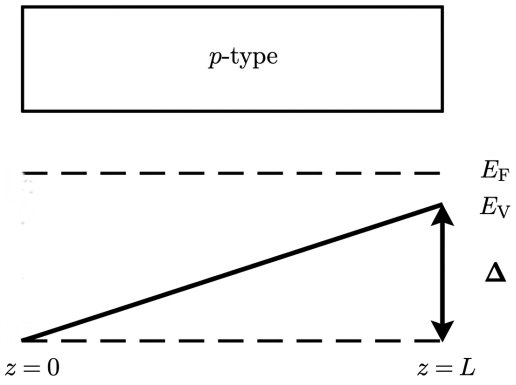


Fig. Q1.21.2

If the charge of an electron is q, then the magnitude of the electric field developed inside this semiconductor bar is

A. $\frac{\Delta}{qL}$ **B.** $\frac{2\Delta}{qL}$ **C.** $\frac{\Delta}{2qL}$ **D.** $\frac{3\Delta}{2qL}$

Q1.20.1 MCQ 2020 1M

A single crystal intrinsic semiconductor is at a temperature of 300 K with effective density of states for holes twice that of electrons. The thermal voltage is 26 mV. The intrinsic Fermi level is shifted from mid-bandgap energy level by

A. 13.45 meV **B.** 18.02 meV
C. 26.90 meV **D.** 9.01 meV

Q1.17.1 NAT 2017 2M

The dependence of drift velocity of electrons on electric field in a semiconductor is shown below. The semiconductor has a uniform electron concentration of $n = 1 \times 10^{16}$ cm^{-3} and electronic charge $q = 1.6 \times 10^{-19}$ C. If a bias of 5 V is applied across a 1 μm region of this semiconductor, the resulting current density in this region, (in kA/cm^2), is ____

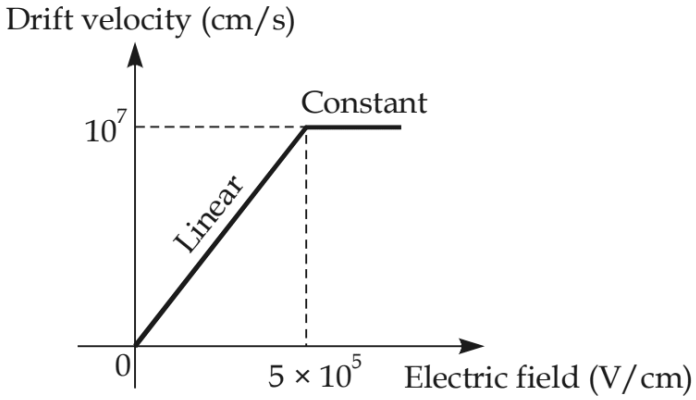


Fig. Q1.17.1

Q1.17.2 NAT 2017 2M

As shown a uniformly doped Silicon (Si) bar of length $L = 0.1$ μm with a donor concentration $N_D = 10^{16}$ cm^{-3} is illuminated at $x = 0$ such that electron and hole pairs are generated at the rate of $G_L = G_{L0}(1 - \frac{x}{L})$, $0 \le x \le L$, where $G_{L0} = 10^{17}$ cm$^{-3}s^{-1}$. Hole lifetime is $10^{-4}s$, electronic charge $q = 1.6 \times 10^{-19}C$, hole diffusion coefficient $D_P = 100$ cm$^2/s$ and low level injection condition prevails. Assuming a linearly decaying steady state excess hole concentration that goes to 0 at $x = L$, the magnitude of the diffusion current density at $x = L/2$, in A/cm^2, is _____.

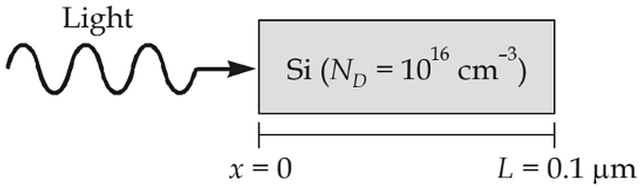


Fig. Q1.17.2

Q1.17.3 MCQ 2017 1M

A bar of Gallium Arsenide ($GaAs$) is doped with Silicon such that the Silicon atoms occupy Gallium and Arsenic sites in the $GaAs$ crystal. Which one of the following statement is true?

A. Silicon atoms act as p-type dopants in Arsenic sites and n-type dopants in Gallium sites

B. Silicon atoms act as n-type dopants in Arsenic sites and p-type dopants in Gallium sites

C. Silicon atoms act as p-type dopants in Arsenic as well as Gallium sites

D. Silicon atoms act as n-type dopants in Arsenic as well as Gallium sites

Q1.16.1 MCQ 2016 2M

Consider a silicon sample at $T = 300$ K, with a uniform donor density $N_d = 5 \times 10^{16}$ cm^{-3} illuminated uniformly such that the optical generation rate is $G_{opt} = 1.5 \times 10^{20}$ cm$^{-3}s^{-1}$ throughout the sample. The incident radiation is turned off at $t = 0$. Assume low-level injection to be valid and ignore surface effects. The carrier lifetimes are $\tau_{p0} = 0.1\mu s$ and $\tau_{n0} = 0.5\mu s$

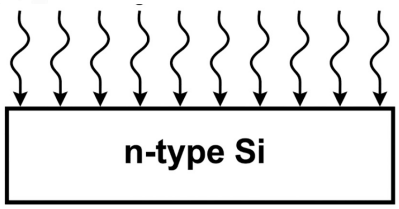


Fig. Q1.16.1

The hole concentration at $t = 0$ and the hole concentration at $t = 0.3$ μs, respectively, are

A. 1.5×10^{13} cm^{-3} and 7.47×10^{11} cm^{-3}

B. 1.5×10^{13} cm^{-3} and 8.23×10^{11} cm^{-3}

C. 7.5×10^{13} cm^{-3} and 3.73×10^{11} cm^{-3}

D. 7.5×10^{13} cm^{-3} and 4.12×10^{11} cm^{-3}

Q1.16.2 NAT 2016 2M

The figure below shows the doping distribution in a p-type semiconductor in log scale. The magnitude of the electric field (in kV/cm) in the semiconductor due to non uniform doping is _____

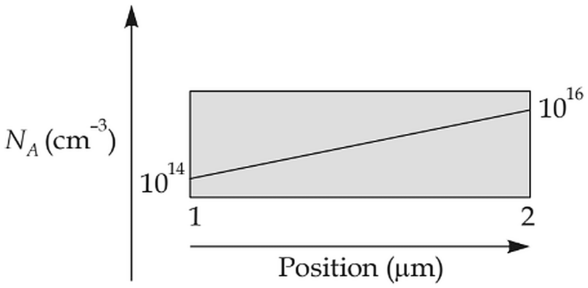


Fig. Q1.16.2

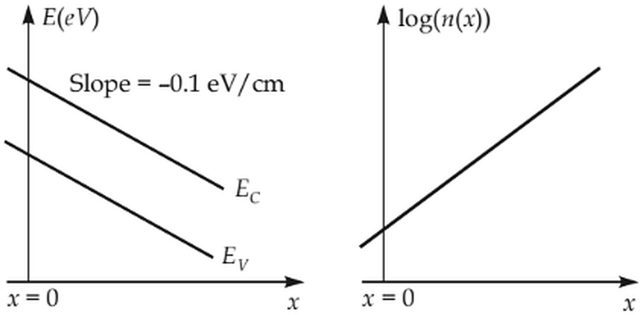


Fig. Q1.15.1

Q1.16.3 MCQ 2016 1M

A small percentage of impurity is added to an intrinsic semiconductor at 300 K. Which one of the following statements is true for the energy band diagram shown in the following figure?

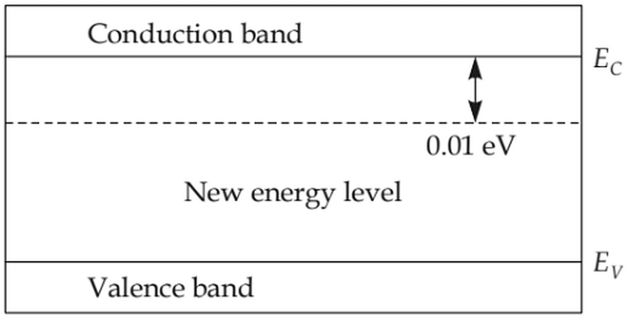


Fig. Q1.16.3

A. Intrinsic semiconductor doped with pentavalent atoms to form n-type semiconductor

B. Intrinsic semiconductor doped with trivalent atoms to form n-type semiconductor

C. Intrinsic semiconductor doped with pentavalent atoms to form p-type semiconductor

D. Intrinsic semiconductor doped with trivalent atoms to form p-type semiconductor

Q1.15.1 MCQ 2015 2M

The energy band diagram and the electron density profile $n(x)$ in a semiconductor are shown in the figures.

Assume that $n(x) = 10^{15} exp(\frac{q\alpha x}{kT})$ cm^{-3}, with $\alpha = 0.1$ V/cm and x expressed in cm. Given $\frac{kT}{q} = 0.026$ V, $D_n = 36$ cm$^2 s^{-1}$, and $\frac{D}{\mu} = \frac{kT}{q}$.

The electron current density (in A/cm^2) at $x = 0$ is

A. -4.4×10^{-2} **B.** -2.2×10^{-2}
C. 0 **D.** 2.2×10^{-2}

Q1.15.2 NAT 2015 2M

A dc voltage of 10 V is applied across an n-type silicon bar having a rectangular cross-section and a length of 1 cm as shown in figure. The donor doping concentration N_D and the mobility of electrons μ_n are 10^{16} cm^{-3} and 1000 cm^2 V^{-1} s^{-1}, respectively. The average time (in μs) taken by the electrons to move from one end of the bar to other end is _____.

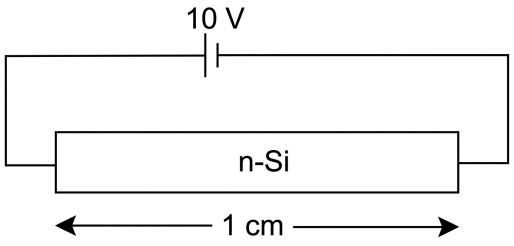


Fig. Q1.15.2

Q1.15.3 NAT 2015 1M

A piece of silicon is doped uniformly with phosphorous with a doping concentration of 10^{16} /cm^3. The expected value of mobility versus doping concentration for silicon assuming full dopant ionization is shown below. The charge of an electron is 1.6×10^{-19}C. The conductivity (in S cm^{-1}) of the silicon sample at 300 K is _____.

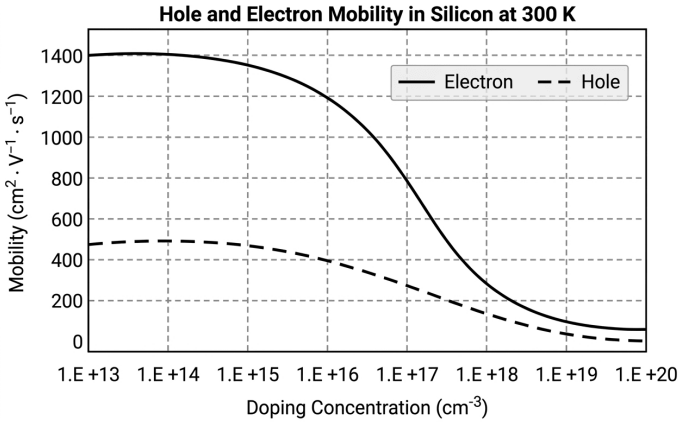


Fig. Q1.15.3

Q1.15.4 NAT 2015 1M

An n-type silicon sample is uniformly illuminated with light which generates 10^{20} electron-hole pairs per cm^3 per second. The minority carrier lifetime in the sample is 1 μs. In the steady state, the hole concentration in the sample is approximately 10^x, where x is an integer. The value of x is _____.

Q1.15.5 NAT 2015 1M

A silicon sample is uniformly doped with donor type impurities with a concentration of 10^{16} / cm^3. The electron and hole mobilities in the sample are 1200 $cm^2/V-s$ and 400 $cm^2/V-s$ respectively. Assume complete ionization of impurities. The charge of an electron is 1.6×10^{-19} C. The resistivity of the sample (in Ω-cm) is _____.

Q1.14.1 MCQ 2014 2M

An N -type semiconductor having uniform doping is biased as shown in the figure.

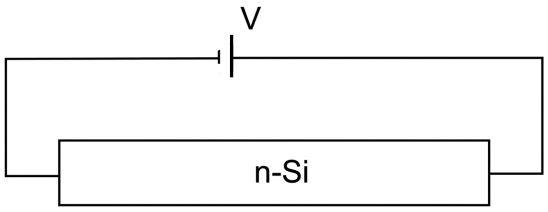
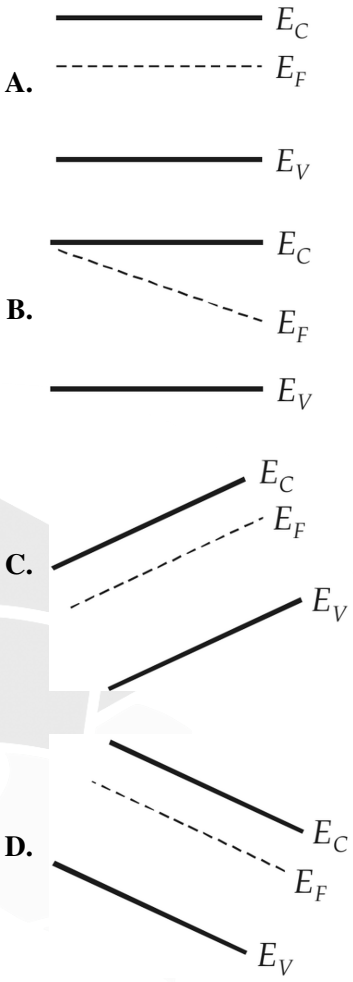


Fig. Q1.14.1

If E_C is the lowest energy level of the conduction band, E_V is the highest energy level of the valence band and E_F is the Fermi level, which one of the following represents the energy band diagram for the biased N-type semiconductor ?



Q1.14.2 NAT 2014 2M

Consider a silicon sample doped with $N_D = 1 \times 10^{15}$ /cm^3 donor atoms. Assume that the intrinsic carrier concentration $n_i = 1.5 \times 10^{10}$ /cm^3. If the sample is additionally doped with $N_A = 1 \times 10^{18}$ /cm^3 acceptor atoms, the approximate number of electrons/cm^3 in the sample, at T = 300 K, will be _____.

Q1.14.3 NAT 2014 1M

The cut-off wavelength (in μ m) of light that can be used for intrinsic excitation of a semiconductor material of bandgap $E_g = 1.1$ eV is _____.

Q1.14.4 MCQ 2014 1M

In the figure, $ln(\rho_i)$ is plotted as a function of $1/T$, where ρ_i is the intrinsic resistivity of silicon, T is the temperature, and the plot is almost linear. The slope of the line can be used to estimate

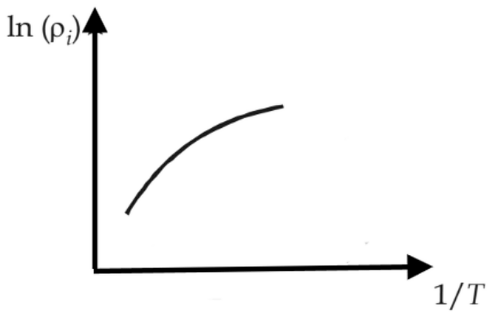


Fig. Q1.14.4

A. band gap energy of silicon (E_g)

B. sum of electron and hole mobility in silicon ($\mu_n + \mu_p$)

C. reciprocal of the sum of electron and hole mobility in silicon $(\mu_n + \mu_p)^{-1}$

D. intrinsic carrier concentration of silicon (n_i)

Q1.14.5 MCQ 2014 1M

At T = 300 K, the band gap and the intrinsic carrier concentration of GaAs are 1.42 eV and $10^6\ cm^{-3}$, respectively. In order to generate electron hole pairs in GaAs, which one of the wavelength (λ_C) ranges of incident radiation, is most suitable ? (Given that : Plank's constant is 6.62×10^{-34}J-s, velocity of light is $3\times10^{10}\ cm/s$ and charge of electron is 1.6×10^{-19}C)

A. $0.42\mu m < \lambda_c < 0.87\mu m$

B. $0.87\mu m < \lambda_c < 1.42\mu m$

C. $1.42\mu m < \lambda_c < 1.62\mu m$

D. $1.62\mu m < \lambda_c < 6.62\mu m$

Q1.14.6 NAT 2014 1M

At T = 300 K, the hole mobility of a semiconductor $\mu_P = 500\ cm^2/Vs$ and $\frac{KT}{q} = 26\ mV$.

The hole diffusion constant D_P in cm^2/s is ____.

Q1.14.7 MCQ 2014 1M

A thin P-type silicon sample is uniformly illuminated with light which generates excess carriers. The recombination rate is directly proportional to

A. the minority carrier mobility

B. the minority carrier recombination lifetime

C. the majority carrier concentration

D. the excess minority carrier concentration

Q1.14.8 NAT 2014 2M

Assume electronic charge $q = 1.6\times10^{-19}\ C$, $kT/q = 25mV$ and electron mobility $\mu_n = 1000\ cm^2/Vs$. If the concentration gradient of electrons injected into a P-type silicon sample is $1\times10^{21}\ /cm^4$, the magnitude of electron diffusion current density (in A/cm^2) is ____.

Q1.14.9 MCQ 2014 1M

A silicon bar is doped with donor impurities $N_D = 2.25\times10^{15}\ atom/cm^3$. Given the intrinsic carrier concentration of silicon at T = 300 K is $n_i = 1.5\times10^{10}\ cm^{-3}$. Assuming complete impurity ionization, the equilibrium electron and hole concentrations are

A. $n_0 = 1.5\times10^{16}\ cm^{-3}, p_0 = 1.5\times10^{5}\ cm^{-3}$

B. $n_0 = 1.5\times10^{10}\ cm^{-3}, p_0 = 1.5\times10^{15}\ cm^{-3}$

C. $n_0 = 2.25\times10^{15}\ cm^{-3}, p_0 = 1.5\times10^{10}\ cm^{-3}$

D. $n_0 = 2.25\times10^{15}\ cm^{-3}, p_0 = 1\times10^{5}\ cm^{-3}$

Q1.11.1 MCQ 2011 1M

Drift current in the semiconductors depends upon

A. only the electric field

B. only the carrier concentration gradient

C. both the electric field and the carrier concentration

D. both the electric field and the carrier concentration gradient

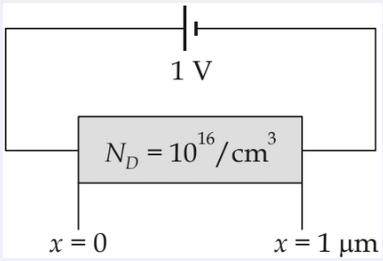
Common Data Questions

The next 2 questions share the data below.

Answer the next two questions using the following data.

The silicon sample with unit cross-sectional areas shown below is in thermal equilibrium. The following information is given:

T = 300 K, electronic charge = 1.6×10^{-19} C, thermal voltage = 26 mV and electron mobility = 1350 cm^2/V-s.



Q1.10.1 MCQ 2010 2M

The magnitude of the electric field at $x = 0.5\ \mu$m is

A. 1 kV/cm **B.** 5 kV/cm
C. 10 kV/cm **D.** 26 kV/cm

Q1.10.2 MCQ 2010 2M

Shared data repeated for this question: electron mobility is 1350 $\text{cm}^2/(\text{V s})$ and electronic charge is 1.6×10^{-19} C. The shared figure gives donor density $10^{16}\ \text{cm}^{-3}$, length 1 μm and applied voltage 1 V.

The magnitude of the electron drift current density at $x = 0.5\ \mu$m is

A. 2.16×10^4 A/cm^2 **B.** 1.08×10^4 A/cm^2
C. 4.32×10^3 A/cm^2 **D.** 6.48×10^2 A/cm^2

Q1.09.1 MCQ 2009 1M

The ratio of the mobility to the diffusion coefficient in a semiconductor has the units

A. V^{-1} **B.** $cm.V^1$ **C.** $V.cm^{-1}$ **D.** $V.s$

Q1.09.2 MCQ 2009 1M

In an n-type silicon crystal at room temperature, which of the following can have a concentration of $4 \times 10^{19}\ cm^{-3}$?

A. Silicon atoms **B.** Holes
C. Dopant atoms **D.** Valence electrons

Q1.08.1 MCQ 2008 1M

Silicon is doped with boron to a concentration of 4×10^{17} atoms/ cm^3. Assume the intrinsic carrier concentration of silicon to be $1.5 \times 10^{10}/cm^3$ and the value of kT/q to be 25 mV at 300 K. Compared to undopped silicon, the fermi level of doped silicon

A. goes down by 0.31 eV

B. goes up by 0.13 eV

C. goes down by 0.427 eV

D. goes up by 0.427 eV

Q1.08.2 MCQ 2008 1M

Which of the following is true?

A. A silicon wafer heavily doped with boron is a p^+ substrate

B. A silicon wafer lightly doped with boron is a p^+ substrate

C. A silicon wafer heavily doped with arsenic is a p^+ substrate

D. A silicon wafer lightly doped with arsenic is a p^+ substrate

Q1.07.1 MCQ 2007 1M

The electron and hole concentrations in an intrinsic semiconductor are n_i per cm^3 at 300 K. Now, if acceptor impurities are introduced with a concentration of N_A per cm^3 (where $N_A >> n_i$), the electron concentration per cm^3 at 300 K will be

A. n_i **B.** $n_i + N_A$ **C.** $N_A - n_i$ **D.** $\frac{n_i^2}{N_A}$

Q1.06.1 MCQ 2006 1M

A heavily doped n-type semiconductor has the following data:

Hole-electron mobility ratio : 0.4
Doping concentration : 4.2×10^8 atoms/m^3
Intrinsic concentration : 1.5×10^4 atoms/m^3

The ratio of conductance of the n-type semiconductor to that of the intrinsic semiconductor of same material and at same temperature is given by

A. 0.00005 **B.** 2000 **C.** 10000 **D.** 20000

Q1.06.2 MCQ 2006 1M

The majority carriers in an n-type semiconductor have an average drift velocity v in a direction perpendicular to a uniform magnetic field B. The electric field E induced due to Hall effect acts in the direction

A. $v \times B$ **B.** $B \times v$
C. along v **D.** opposite to v

Q1.06.3 MCQ 2006 1M

Under low level injection assumption, the injected minority carrier current for an extrinsic semiconductor is essentially the

A. Diffusion current **B.** Drift current
C. Recombination current **D.** Induced current

Q1.06.4 MCQ 2006 1M

The concentration of minority carriers in an extrinsic semiconductor under equilibrium is

A. Directly proportional to doping concentration

B. Inversely proportional to the doping concentration

C. Directly proportional to the intrinsic concentration

D. Inversely proportional to the intrinsic concentration

Q1.05.1 MCQ 2005 1M

A Silicon sample A is doped with 10^{18} atoms/cm^3 of boron. Another sample B of identical dimension is doped with 10^{18} atoms/cm^3 phosphorus. The ratio of electron to hole mobility is 3. The ratio of conductivity of the sample A to B is

A. 3 **B.** $\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{2}$

Q1.05.2 MCQ 2005 1M

The primary reason for the widespread use of Silicon in semiconductor device technology is

A. abundance of Silicon on the surface of the Earth

B. larger bandgap of Silicon in comparison to Germanium

C. favorable properties of Silicon dioxide (SiO2)

D. lower melting point

Q1.05.3 MCQ 2005 1M

The bandgap of Silicon at room temperature is

A. 1.3 eV **B.** 0.7 eV **C.** 1.1 eV **D.** 1.4 eV

Q1.04.1 MCQ 2004 1M

The longest wavelength that can be absorbed by silicon, which has the bandgap of 1.12 eV, is 1.1 μm. If the longest wavelength that can be absorbed by another material is 0.87 μm, then bandgap of this material is

A. 1.416 eV **B.** 0.886 eV
C. 0.854 eV **D.** 0.706 eV

Q1.04.2 MCQ 2004 1M

The resistivity of a uniformly doped n-type silicon sample is 0.5 Ωcm. If the electron mobility (μ_n) is 1250 cm^2/Vsec and the charge of an electron is 1.6×10^{-19} Coulomb, the donor impurity concentration (N_D) in the sample is

A. $2 \times 10^{16}/cm^3$ **B.** $1 \times 10^{16}/cm^3$
C. $2.5 \times 10^{15}/cm^3$ **D.** $5 \times 10^{15}/cm^3$

Q1.04.3 MCQ 2004 1M

The impurity commonly used for realizing the base region of a silicon n - p - n transistor is

A. Gallium **B.** Indium
C. Boron **D.** Phosphorus

Q1.03.1 MCQ 2003 1M

The electron concentration in a sample of uniformly doped n-type silicon at 300 K varies linearly from $10^{17}/cm^3$ at $x = 0$ to $6 \times 10^{16}/cm^3$ at $x = 2\mu m$. Assume a situation that electrons are supplied to keep this concentration gradient constant with time. If electronic charge is 1.6×10^{-19} coulomb and the diffusion constant $D_n = 35\ cm^2/s$, the current density in the silicon, if no electric field is present, is

A. zero **B.** $-112A/cm^2$
C. $+1120A/cm^2$ **D.** $-1120A/cm^2$

Q1.03.2 MCQ 2003 1M

An n-type silicon bar 0.1 cm long and 100 μm^2 in cross-sectional area has a majority carrier concentration of $5 \times 10^{20}/m^3$ and the carrier mobility is 0.13 $m^2/V-s$ at 300 K. If the charge of an electron is 1.6×10^{-19} coulomb, then the resistance of the bar is

A. 10^6 Ohm **B.** 10^4 Ohm
C. 10^{-1} Ohm **D.** 10^{-4} Ohm

Q1.02.1 MCQ 2002 1M

The intrinsic carrier concentration of silicon sample at 300 K is $1.5 \times 10^{16}/m^3$. If after doping, the number of majority carriers is $5 \times 10^{20}/m^3$, the minority carrier density is

A. $4.50 \times 10^{11}/m^3$ **B.** $3.333 \times 10^4/m^3$
C. $5.00 \times 10^{20}/m^3$ **D.** $3.00 \times 10^{-5}/m^3$

Q1.98.1 MCQ 1998 1M

The units of $\frac{q}{kT}$ are

A. V **B.** V^{-1} **C.** J **D.** J/K

Q1.95.1 MCQ 1995 1M

A small concentration of minority carriers is injected into a homogeneous semiconductor crystal at one point. An electric field of 10 V/cm is applied across the crystal and this moves the minority carries a distance of 1 cm in 20 μsec. The mobility (in cm^2/V-sec) will be

A. 1, 000 **B.** 2, 000 **C.** 5, 000 **D.** 500, 000

Q1.95.2 MCQ 1995 1M

In a p-type Si simple the hole concentration is $2.25 \times 10^{15}/cm^3$. The intrinsic carrier concentration is $1.5 \times 10^{10}/cm^3$ the electron concentration is

A. zero **B.** $10^{10}/cm^3$
C. $10^5/cm^3$ **D.** $1.5 \times 10^{25}/cm^3$

Q1.95.3 MCQ 1995 1M

The probability that an electron in a metal occupies the Fermi level, at any temperature ($T > 0$ K)

A. 0 **B.** 1 **C.** 0.5 **D.** 1.0

Q1.95.4 MCQ 1995 1M

The drift velocity of electrons, in silicon

A. is proportional to the electric field for all values of electric field.

B. is independent of the electric field.

C. increases at low values of electric field and decreases at high values of electric field exhibiting negative differential resistance.

D. increases linearly with electric field at low values of electric field and gradually saturates at higher values of electric field.

Q1.94.1 MCQ 1994 1M

A p-type silicon sample has a higher conductivity compared to an n-type silicon sample having the same dopant concentration. (True/False) _____

Q1.92.1 MCQ 1992 2M

A semiconductor is irradiated with light such that carriers are uniformly generated throughout its volume. The semiconductor is n-type with $N_D = 10^{19}/cm^3$. If the excess electron concentration in the steady-state is $\Delta n = 10^{15}/cm^3$ and if $\tau_p = 10\,\mu$sec. (minority carriers life time) the generation rate due to irradiation

A. is 10^{20} e-h pairs/cm^3/s

B. is 10^{24} e-h pairs/cm^3/s

C. is 10^{10} e-h pairs/cm^3/s

D. cannot be determined, the given data is insufficient

Q1.91.1 MCQ 1991 2M

A silicon sample is uniformly doped with 10^{16} phosphorus atoms/cm^3 and 2×10^{16} boron atoms/cm^3. If all the dopants are fully ionized, the material is

A. n-type with carrier concentration of $10^{16}/cm^3$.

B. p-type with carrier concentration of $10^{16}/cm^3$.

C. p-type with carrier of $2 \times 10^{16}/cm^3$.

D. n-type with a carrier concentration of $2 \times 10^{16}/cm^3$.

Q1.90.1 MSQ 1990 2M

Under high electric fields, in a semiconductor with increasing electric field,

A. the mobility of charge carriers decreases.

B. the mobility of charge carriers increases.

C. the velocity of the charge carriers saturates.

D. the velocity of the charge carriers increases.

Q1.89.1 MCQ 1989 2M

The concentration of ionized acceptors and donors in a semiconductor are N_A, N_D respectively. If $N_A > N_D$ and n_i is the intrinsic concentration, the position of the Fermi level with respect to the intrinsic level depends on

A. $N_A - N_D$ **B.** $N_A + N_D$ **C.** $\frac{N_A N_D}{n_i^2}$ **D.** n_i

Q1.89.2 MCQ 1989 2M

Due to illumination by light, the electron and hole concentrations in a heavily doped N type semiconductor increases by Δn and Δp respectively if n_i is the intrinsic concentration then,

A. $\Delta n < \Delta p$ **B.** $\Delta n > \Delta p$
C. $\Delta n = \Delta p$ **D.** $\Delta n \times \Delta p = n_i^2$

Q1.87.1 MCQ 1987 2M

Consider two energy levels: E_1, E ev above the Fermi level and E_2, E ev below the Fermi level. P_1 and P_2 are respectively the probability of E_1 being occupied by an electron and E_2 being empty. Then,

A. $P_1 > P_2$ **B.** $P_1 = P_2$
C. $P_1 < P_2$ **D.** P_1 & P_2 depend on number of free electrons

Q1.87.2 MCQ 1987 2M

In an intrinsic semiconductor the free electron concentration depends on

A. effective mass of electrons only

B. effective mass of holes only

C. temperature of the semiconductor

D. width of the forbidden energy band of the semiconductor

Q1.87.3 MCQ 1987 2M

According to the Einstein relation, for any semiconductor the ratio of diffusion constant to mobility of carriers

A. depends upon the temperature of the semiconductor.

B. depends upon the type of the semiconductor.

C. varies with life time of the semiconductor.

D. is a universal constant.

Q1.87.4 MCQ 1987 2M

Direct band gap semiconductors

A. exhibit short carrier life time and they are used for fabricating BJT's.

B. exhibit long carrier life time and they are used for fabricating BJT's.

C. exhibit short carrier life time and they are used for fabricating lasers.

D. exhibit long carrier life time and they are used for fabricating lasers.

Answer Key

Q1.26.1 -36.50 to -34.50 OR 34.50 to 36.50	Q1.25.1 A,D	Q1.25.2 B
Q1.25.3 B	Q1.24.1 A, C, D	Q1.24.2 A
Q1.24.3 0.17 to 0.19	Q1.23.1 A	Q1.23.2 A
Q1.23.3 2.20 to 2.30	Q1.23.4 60.00 to 70.00	Q1.22.1 A, C
Q1.22.2 A	Q1.22.3 C	Q1.22.4 0.57 to 0.61
Q1.21.1 C	Q1.21.2 A	Q1.20.1 D

Q1.17.1 1.5 to 1.7	Q1.17.2 15.9 to 16.1	Q1.17.3 A
Q1.16.1 A	Q1.16.2 1.1 to 1.25	Q1.16.3 A
Q1.15.1 C	Q1.15.2 95 to 105	Q1.15.3 1.8 to 2
Q1.15.4 14	Q1.15.5 0.50 to 0.54	Q1.14.1 D
Q1.14.2 224.9 to 225.1	Q1.14.3 1.12 to 1.14	Q1.14.4 A
Q1.14.5 A	Q1.14.6 12.9 to 13.1	Q1.14.7 D

Q1.14.8 3990 to 4010	Q1.14.9 D	Q1.11.1 C
Q1.10.1 C	Q1.10.2 A	Q1.09.1 A
Q1.09.2 C	Q1.08.1 C	Q1.08.2 A
Q1.07.1 D	Q1.06.1 D	Q1.06.2 B
Q1.06.3 A	Q1.06.4 B	Q1.05.1 B
Q1.05.2 C	Q1.05.3 C	Q1.04.1 A

Q1.04.2 B	Q1.04.3 C	Q1.03.1 D
Q1.03.2 A	Q1.02.1 A	Q1.98.1 B
Q1.95.1 C	Q1.95.2 C	Q1.95.3 C
Q1.95.4 D	Q1.94.1 False	Q1.92.1 A
Q1.91.1 B	Q1.90.1 A,C	Q1.89.1 A
Q1.89.2 C	Q1.87.1 B	Q1.87.2 C
Q1.87.3 A	Q1.87.4 C	

PrepFusion

UNIT

II

PN Junctions

Topics

1. Basic Semiconductor Junctions
2. PN Junction Under Equilibrium
3. PN Junction Under Biasing

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 57	**MCQ:** 43	**MSQ:** 2	**NAT:** 12
1M: 30	**2M:** 27	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2013)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

Consider carrier transport in a Zener diode in the breakdown region. Which is the dominant transport mechanism for current flow in this case?

A. Drift **B.** Diffusion
C. Tunneling **D.** Ballistic transport

Q2.26.2 MSQ 2026 1M

Consider a p-n junction diode when it is forward biased with 2 V. Which of the following is/are the correct magnitude(s) of the energy difference between quasi Fermi-levels, E_{fn} in the n-side and E_{fp} in the p-side?

- ☐ **A.** 2 eV
- ☐ **B.** 1 eV
- ☐ **C.** 2 V
- ☐ **D.** 1 V

Q2.25.1 NAT 2025 2M

An ideal p-n junction germanium diode has a reverse saturation current of 10 μA at 300 K. The voltage (in Volts, rounded off to two decimal places) to be applied across the junction to get a forward bias current of 100 mA at 300 K is _____.

(Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}$J/K and the charge of an electron $e = 1.6 \times 10^{-19}$ C.)

Q2.21.1 NAT 2021 2M

A silicon P-N junction is shown in the figure. The doping in the P region is 5×10^{16} cm^{-3} and doping in the N region is 10×10^{16} cm^{-3}. The parameters given are:

- Built-in voltage $\Phi_{bi} = 0.8$ V.
- Electron charge $q = 1.6 \times 10^{-19}$ C.
- Vacuum permittivity $\epsilon_0 = 8.85 \times 10^{-12}$ F/m.
- Relative permittivity of silicon $\epsilon_{Si} = 12$.

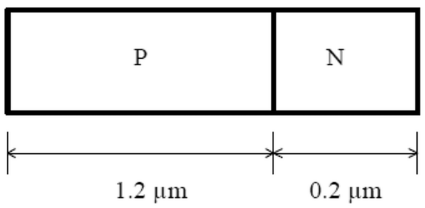


Fig. Q2.21.1

The magnitude of reverse bias voltage that would completely deplete one of the two regions (P or N) prior to the other (rounded off to one decimal place) is _____ V.

Q2.20.1 MCQ 2020 1M

Consider the recombination process via bulk traps in a forward biased pn homojunction diode. The maximum recombination rate is U_{max}. If the electron and the hole capture cross-section are equal, which one of the following is False?

A. With all other parameters unchanged, U_{max} decreases if the intrinsic carrier density is reduced.

B. U_{max} occurs at the edges of the depletion region in the device.

C. U_{max} depends exponentially on the applied bias.

D. With all other parameters unchanged, U_{max} increases if the thermal velocity of the carriers increases.

Q2.20.2 MCQ 2020 2M

A one-sided abrupt pn junction diode has a depletion capacitance C_D of 50 pF at a reverse bias of 0.2 V. The plot of $1/C_D^2$ versus the applied voltage V for this diode is a straight line as shown in the figure below. The slope of the plot is _____ $\times 10^{20}\ F^{-2}\ V^{-1}$.

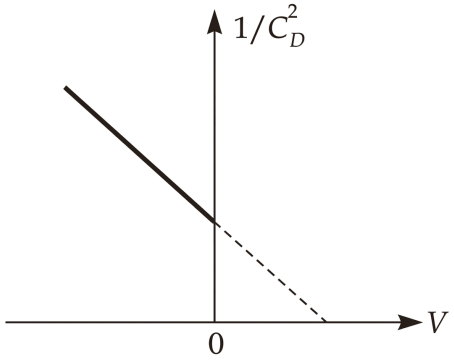


Fig. Q2.20.2

A. -5.7 **B.** -3.8 **C.** -1.2 **D.** -0.4

Q2.19.1 NAT 2019 2M

In an ideal pn junction with an ideality factor of 1 at $T = 300\ K$, the magnitude of the reverse-bias voltage required to reach 75% of its reverse saturation current, rounded off to 2 decimal places, is _____ mV. [$k = 1.38 \times 10^{-23}\ JK^{-1}$, $h = 6.625 \times 10^{-34}\ Js$, $q = 1.602 \times 10^{-19}\ C$]

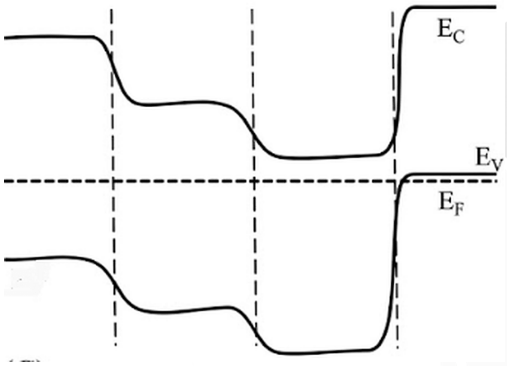
Q2.19.2 MCQ 2019 1M

Which one of the following options describes correctly the equilibrium band diagram at $T = 300\ K$ of a Silicon $pnp^{+}p^{++}$ configuration shown in the figure?

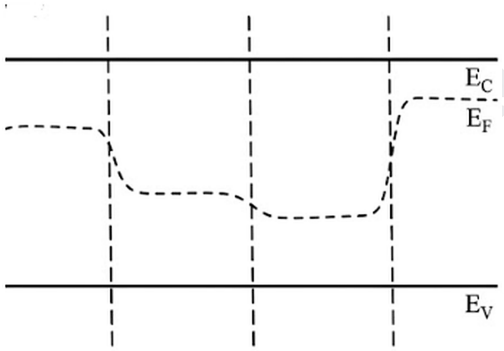


Fig. Q2.19.2

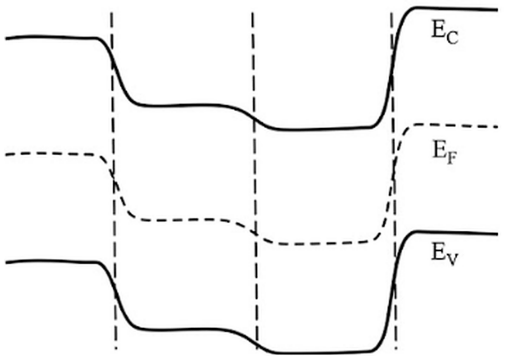
A.



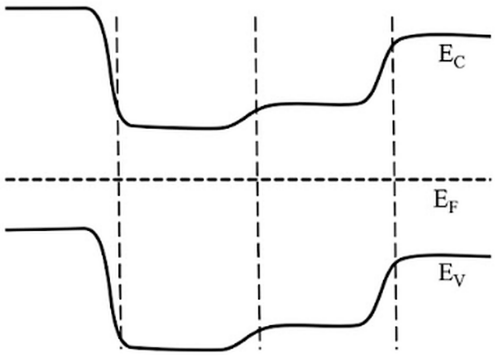
B.



C.



D.



Q2.18.1 NAT 2018 2M

A junction is made between p^{-} Si with doping density $N_{A1} = 10^{15} cm^{-3}$ and p Si with doping density $N_{A2} = 10^{17} cm^{-3}$. Given: Boltzmann constant $k = 1.38 \times 10^{-23} JK^{-1}$, electronic charge $q = 1.6 \times 10^{-19} C$. Assume 100% acceptor ionization. At room temperature ($T = 300K$), the magnitude of the built-in potential (in volts, correct to two decimal places) across this junction will be ____.

Q2.18.2 NAT 2018 1M

A p-n step junction diode with a contact potential of $0.65V$ has a depletion width of $1\mu m$ at equilibrium. The forward voltage (in volts, correct to two decimal places) at which this width reduces to $0.6\mu m$ is ____.

Q2.18.3 MCQ 2018 1M

In a p-n junction diode at equilibrium, which one of the following statements is NOT TRUE?

A. The hole and electron diffusion current components are in the same direction.

B. The hole and electron drift current components are in the same direction.

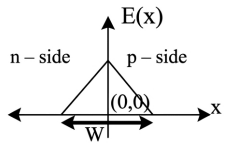
C. On an average, holes and electrons drift in opposite direction.

D. On an average, electrons drift and diffuse in the same direction.

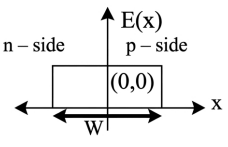
Q2.17.1 MCQ 2017 2M

An abrupt pn junction (located at $x = 0$) is uniformly doped on both p and n sides. The width of the depletion region is W and the electric field variation in the x-direction is $E(x)$. Which of the following figures represents the electric field profile near the pn junction?

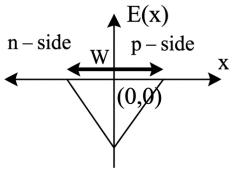
(A)



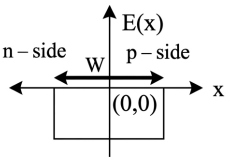
(B)



(C)



(D)



A. A **B.** B **C.** C **D.** D

Q2.17.2 NAT 2017 2M

As shown, two Silicon (Si) abrupt p-n junction diodes are fabricated with uniform donor doping concentration of $N_{D1} = 10^{14}cm^{-3}$ and $N_{D2} = 10^{16}cm^{-3}$ in the n-regions of the diodes, and uniform acceptor doping concentration of $N_{A1} = 10^{14}cm^{-3}$ and $N_{A2} = 10^{16}cm^{-3}$ in the p regions of the diodes, respectively. Assuming that the reverse bias voltage is >> built-in potentials of the diodes, the ratio $\frac{C_2}{C_1}$ of their reverse bias capacitances for the same applied reverse bias, is ____.

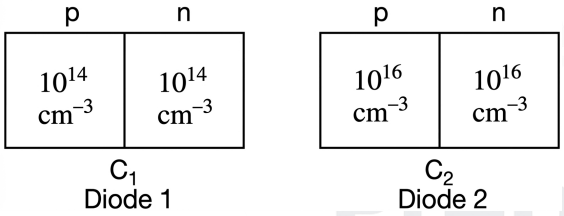


Fig. Q2.17.2

Q2.17.3 MCQ 2017 1M

An $n^+ - n$ Silicon device is fabricated with uniform and non-degenerate donor doping concentrations of $N_{D1} = 1 \times 10^{18}cm^{-3}$ and $N_{D2} = 1 \times 10^{15}cm^{-3}$ corresponding to the n^+ and n regions respectively. At the operational temperature T, assume complete impurity ionization, $kT/q = 25mV$, and intrinsic carrier concentration to be $n_i = 1 \times 10^{10}cm^{-3}$. What is the magnitude of the built-in potential of this device?

A. 0.748V **B.** 0.460V **C.** 0.288V **D.** 0.173V

Q2.16.1 MCQ 2016 1M

The I-V characteristics of three types of diodes at the room temperature, made of semiconductors X, Y and Z, are shown in the figure. Assume that the diodes are uniformly doped and identical in all respects except their materials. If E_{gX}, E_{gY} and E_{gZ} are the band gaps of X, Y and Z, respectively, then

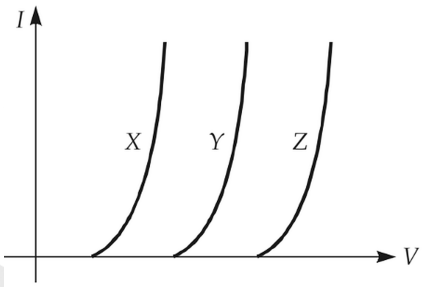


Fig. Q2.16.1

A. $E_{gX} > E_{gY} > E_{gZ}$

B. $E_{gX} = E_{gY} = E_{gZ}$

C. $E_{gX} < E_{gY} < E_{gZ}$

D. no relationship among these band gaps exists.

Q2.16.2 NAT 2016 2M

Consider a region of silicon devoid of electrons and holes, with an ionized donor density of $N_d^+ = 10^{17}cm^{-3}$. The electric field at x= 0 is $0V/cm$ and the electric field at x=L is $50kV/cm$ in the positive x direction. Assume that the electric field is zero in the y and z directions at all points.

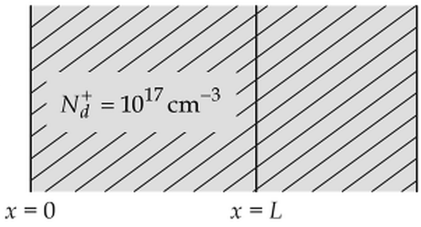


Fig. Q2.16.2

Given $q = 1.6 \times 10^{-19}coulomb$, $\epsilon_0 = 8.85 \times 10^{-14}F/cm$, $\epsilon_r = 11.7$ for silicon, the value of L in nm is _____.

Q2.16.3 MCQ 2016 2M

Consider avalanche breakdown in a silicon p^+n junction. The n-region is uniformly doped with a donor density N_D. Assume that breakdown occurs when the magnitude of the electric field at any point in the device becomes equal to the critical field E_{crit}. Assume E_{crit} to be independent of N_D. If the built-in voltage of the p^+n junction is much smaller than the breakdown voltage, V_{BR}, the relationship between V_{BR} and N_D is

A. $V_{BR} \times \sqrt{N_D} = constant$

B. $N_D \times \sqrt{V_{BR}} = constant$

C. $N_D \times V_{BR} = constant$

D. $N_D / V_{BR} = constant$

Q2.16.4 NAT 2016 2M

Consider a silicon p-n junction with a uniform acceptor doping concentration of $10^{17} cm^{-3}$ on the p-side and a uniform donor doping concentration of $10^{16} cm^{-3}$ on the n-side. No external voltage is applied to the diode. Given: $kT/q = 26mV$, $n_i = 1.5 \times 10^{10} cm^{-3}$, $\varepsilon_{Si} = 12\varepsilon_0$, $\varepsilon_0 = 8.85 \times 10^{-14} F/m$, and $q = 1.6 \times 10^{-19} C$. The charge per unit junction area ($nC\ cm^{-2}$) in the depletion region on the p-side is ____.

Q2.15.1 MCQ 2015 2M

The electric field profile in the depletion region of a p-n junction in equilibrium is shown in the figure. Which one of the following statements is NOT TRUE?

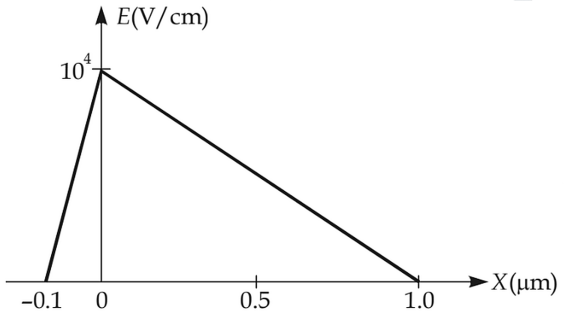


Fig. Q2.15.1

A. The left side of the junction is n-type and the right side is p-type

B. Both the n-type and p-type depletion regions are uniformly doped

C. The potential difference across the depletion region is 700 mV

D. If the p-type region has a doping concentration of 10^{15} cm^{-3}, then the doping concentration in the n-type region will be 10^{16} cm^{-3}

Q2.15.2 NAT 2015 2M

For a silicon diode with long P and N regions, the accepter and donor impurity concentrations are 1×10^{17} cm^{-3} and 1×10^{15} cm^{-3}, respectively. The lifetimes of electrons in P region and holes in N region are both 100 μs. The electron and hole diffusion coefficients are 49 cm^2/s and 36 cm^2/s, respectively. Assume $kT/q = 26$ mV, the intrinsic carrier concentration is 1×10^{10} cm^{-3}, and $q = 1.6 \times 10^{-19}$ C. When a forward voltage of 208 mV is applied across the diode, the hole current density (in nA/cm^2) injected from P region to N region is ____.

Q2.15.3 NAT 2015 2M

The built-in potential of an abrupt p-n junction is 0.75 V. If its junction capacitance (C_J) at a reverse bias (V_R) of 1.25 V is 5 pF, the value of C_J (in pF) when $V_R = 7.25$ V is ____.

Q2.15.4 MCQ 2015 1M

A region of negative differential resistance is observed in the current voltage characteristics of a silicon PN junction if

A. both the P-region and the N-region are heavily doped

B. the N-region is heavily doped compared to the P-region

C. the P-region is heavily doped compared to the N-region

D. an intrinsic silicon region is inserted between the P-region and the N-region

Q2.14.1 MCQ 2014 2M

The donor and accepter impurities in an abrupt junction silicon diode are 1×10^{16} cm^{-3} and 5×10^{18} cm^{-3}, respectively. Assume that the intrinsic carrier concentration in silicon $n_i = 1.5 \times 10^{10}$ cm^{-3} at 300 K, $\frac{kT}{q} = 26$ mV and the permittivity of silicon $\varepsilon_{si} = 1.04 \times 10^{-12}$ F/cm. The built-in potential and the depletion width of the diode under thermal equilibrium conditions, respectively, are

A. 0.7 V and 1×10^{-4} cm

B. 0.86 V and 1×10^{-4} cm

C. 0.7 V and 3.3×10^{-5} cm

D. 0.86 V and 3.3×10^{-5} cm

Q2.14.2 NAT 2014 2M

Consider an abrupt PN junction (at $T = 300\,\text{K}$) shown in the figure. The depletion region width X_n on the N-side of the junction is $0.2\,\mu\text{m}$ and the permittivity of silicon (ε_{si}) is $1.044 \times 10^{-12}\,\text{F/cm}$. At the junction, the approximate value of the peak electric field (in kV/cm) is ____.

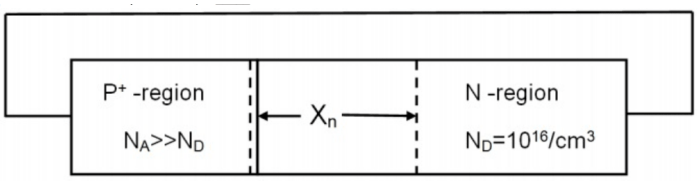


Fig. Q2.14.2

Q2.14.3 MCQ 2014 2M

When a silicon diode having a doping concentration of $N_A = 9 \times 10^{16}\ cm^{-3}$ on p -side and $N_D = 1 \times 10^{16}\ cm^{-3}$ on n-side is reverse biased, the total depletion width is found to be 3 μm. Given that the permittivity of silicon is 1.04×10^{-12} F/cm, the depletion width on the p-side and the maximum electric field in the depletion region, respectively, are

A. $2.7\mu m$ and $2.3 * 10^5\ V/cm$

B. $0.3\mu m$ and $4.15 * 10^5\ V/cm$

C. $0.3\mu m$ and $0.42 * 10^5\ V/cm$

D. $2.1\mu m$ and $0.42 * 10^5\ V/cm$

Q2.14.4 MCQ 2014 2M

The doping concentrations on the p-side and n-side of a silicon diode are $1 \times 10^{16}\ cm^{-3}$ and $1 \times 10^{17}\ cm^{-3}$, respectively. A forward bias of 0.3 V is applied to the diode. At T=300K, the intrinsic carrier concentration of silicon $n_i = 1.5 \times 10^{10}\ cm^{-3}$ and $\frac{KT}{q} = 26\,mV$. The electron concentration at the edge of the depletion region on the p-side is

A. $2.3 \times 10^9\ cm^{-3}$ **B.** $1 \times 10^{16}\ cm^{-3}$
C. $1 \times 10^{17}\ cm^{-3}$ **D.** $2.25 \times 10^9\ cm^{-3}$

Q2.13.1 MCQ 2013 1M

In a forward biased pn junction diode, the sequence of events that best describes the mechanism of current flow is

A. injection, and subsequent diffusion and recombination of minority carriers

B. injection, and subsequent drift and generation of minority carriers

C. extraction, and subsequent diffusion and generation of minority carriers

D. extraction, and subsequent drift and recombination of minority carriers

Q2.11.1 MCQ 2011 1M

A Zener diode, when used in voltage stabilization circuits, is biased in

A. reverse bias region below the breakdown voltage

B. reverse breakdown region

C. forward bias region

D. forward bias constant current mode

Q2.11.2 MCQ 2011 1M

A silicon PN junction is forward biased with a constant current at room temperature. When the temperature is increased by 10° C, the forward bias voltage across the PN junction

A. increases by 60 mV **B.** decreases by 60 mV
C. increases by 25 mV **D.** decreases by 25 mV

Q2.10.1 MCQ 2010 1M

Compared to a p-n junction with $N_A = N_D = 10^{14}/\text{cm}^3$, which one of the following statements is TRUE for a p-n junction with $N_A = N_D = 10^{20}/\text{cm}^3$?

A. Reverse breakdown voltage is lower and depletion capacitance is lower

B. Reverse breakdown voltage is higher and depletion capacitance is lower

C. Reverse breakdown voltage is lower and depletion capacitance is higher

D. Reverse breakdown voltage is higher and depletion capacitance is higher

Common Data Questions

The next 2 questions share the data below.

Answer the next two questions using the following data.

Consider a silicon p-n junction at room temperature having the following parameters:

Doping on the n-side = 1×10^{17} cm^{-3}
Depletion width on the n-side = 0.1 μm
Depletion width on the p-side = 1.0 μm
Intrinsic carrier concentration = 1.4×10^{10} cm^{-3}
Thermal voltage = 26 mV
Permittivity of free space = 8.85×10^{-14} Fcm^{-1}
Dielectric constant of silicon = 12

Q2.09.1 MCQ 2009 1M

The peak electric field in the device is

A. 0.15 MVcm^{-1}, directed from p-region to n-region

B. 0.15 MVcm^{-1}, directed from n-region to p-region

C. 1.80 MVcm^{-1}, directed from p-region to n-region

D. 1.80 MVcm^{-1}, directed from n-region to p-region

Q2.09.2 MCQ 2009 1M

Shared data repeated for this question: donor doping is 10^{17} cm^{-3}; depletion widths on the n and p sides are 0.1 and 1.0 μm, respectively. Intrinsic carrier density is 1.4×10^{10} cm^{-3} and thermal voltage is 26 mV. These are the original common-data values.

The built-in potential of the junction

A. is 0.70 V

B. is 0.76 V

C. is 0.82 V

D. Cannot be estimated from the data given

Q2.08.1 MCQ 2008 1M

Consider the following assertions.
S1 : For Zener effect to occur, a very abrupt junction is required.
S2 : For quantum tunneling to occur, a very narrow energy barrier is required.
Which of the following is correct ?

A. Only $S2$ is true

B. $S1$ and $S2$ are both true but $S2$ is not a reason for $S1$

C. $S1$ and $S2$ are both true but $S2$ is a reason for $S1$

D. Both $S1$ and $S2$ are false

Q2.08.2 MCQ 2008 1M

Which of the following is NOT associated with a p-n junction ?

A. Junction Capacitance

B. Charge Storage Capacitance

C. Depletion Capacitance

D. Channel Length Modulation

Q2.07.1 MCQ 2007 1M

A p^+n junction has a built-in potential of 0.8 V. The depletion layer width a reverse bias of 1.2 V is 2 μm . For a reverse bias of 7.2 V, the depletion layer width will be

A. 4 μm **B.** 4.9 μm **C.** 8 μm **D.** 12 μm

Q2.07.2 MCQ 2007 1M

In a p^+n junction diode under reverse bias the magnitude of electric field is maximum at

A. the edge of the depletion region on the p-side

B. the edge of the depletion region on the n-side

C. the p^+n junction

D. the centre of the depletion region on the n-side

Q2.05.1 MCQ 2005 1M

A Silicon PN junction diode under reverse bias has depletion region of width 10 μm. The relative permittivity of Silicon, $\varepsilon_r = 11.7$ and the permittivity of free space $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m . The depletion capacitance of the diode per square meter is

A. 100 μF **B.** 10 μF **C.** 1 μF **D.** 20 μF

Q2.05.2 MCQ 2005 1M

A Silicon PN junction at a temperature of 20°C has a reverse saturation current of 10 pico-Amperes (pA). The reverse saturation current at 40°C for the same bias is approximately

A. 30pA **B.** 40pA **C.** 50pA **D.** 60pA

Q2.04.1 MCQ 2004 1M

Consider an abrupt p-n junction. Let V_{bi} be the built-in potential of this junction and V_R be the applied reverse bias. If the junction capacitance (C_1) is 1 pF for $V_{bi} + V_R = 1$ V, then for $V_{bi} + V_R = 4$ V, C_j will be

A. 4 pF **B.** 2 pF **C.** 0.25 pF **D.** 0.5 pF

Q2.04.2 MCQ 2004 1M

In an abrupt p-n junction, the doping concentrations on the p-side and n-side are $N_A = 9 \times 10^{16}/\text{cm}^3$ and $N_D = 1 \times 10^{16}/\text{cm}^3$ respectively. The p-n junction is reverse biased and the total depletion width is 3 μm. The depletion width on the p-side is

A. 2.7 μm **B.** 0.3 μm **C.** 2.25 μm **D.** 0.75 μm

Q2.03.1 MCQ 2003 1M

At 300 K, for a diode current of 1 mA, a certain germanium diode requires a forward bias of 0.1435 V, whereas a certain silicon diode requires a forward bias of 0.718 V. Under the conditions state above, the closest approximation of the ratio of reverse saturation current in germanium diode to that in silicon diode is

A. 1 **B.** 5
C. 4×10^3 **D.** 8×10^3

Q2.02.1 MCQ 2002 1M

In Figure shown below, a silicon is carrying a constant current of 1 mA. When the temperature of the diode is 20°C, V_D is found to be 700 mV. If the temperature rises to 40°C, V_D becomes approximately equal to

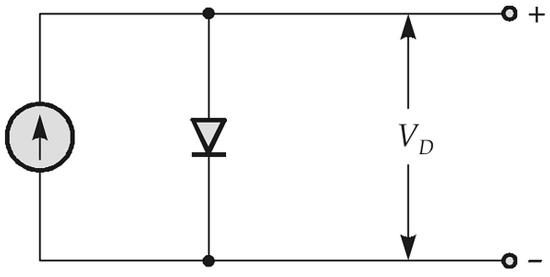


Fig. Q2.02.1

A. 740 mV **B.** 660 mV **C.** 680 mV **D.** 700 mV

Q2.98.1 MCQ 1998 2M

A pn-junction in series with a 100 Ω resistor, is forwarded biased. So that a current of 100 mA flows. If the voltage across this combination is instantaneously reversed at $t = 0$ current through diodes is approximately given by

A. 0 mA **B.** –100 mA **C.** 100 mA **D.** 50 mA

Q2.98.2 MCQ 1998 1M

The static characteristic of an adequately forward biased p-n junction is a straight line, if the plot is of

A. $\log I$ vs $\log V$ **B.** $\log I$ vs V
C. I vs $\log V$ **D.** I vs V

Q2.95.1 MCQ 1995 1M

The depletion capacitance, C_J of an abruptly p-n junction with constant doping on either side varies with R.B. V_R as,

A. $C_J \propto V_R$ **B.** $C_J \propto V_R^{-1}$
C. $C_J \propto V_R^{-1/2}$ **D.** $C_J \propto V_R^{-1/3}$

Q2.95.2 MCQ 1995 1M

A Zener diode works on the principle of

A. tunneling of charge carriers across the junction.

B. thermionic emission.

C. diffusion of charge carriers across the junction.

D. hopping of charge carriers across the junction.

Q2.95.3 MCQ 1995 1M

The diffusion potential across a p-n junction

A. decreases with increasing doping concentration.

B. increases with decreasing bandgap.

C. does not depend on doping concentrations.

D. increases with increase in doping concentration.

Q2.93.1 MSQ 1993 2M

The built-in potential (diffusion potential) in a p-n junction

A. is equal to the difference in the Fermi-level of the two sides, expressed in volts.

B. increases with the increase in the doping levels of the two sides.

C. increases with the increase in temperature.

D. is equal to the average of the Fermi-level of the two sides.

Q2.91.1 NAT 1991 2M

Referring to the below figure the switch 'S' is in position 1 initially and steady-state condition exist from time $t = 0$ to $t = t_0$, the switch is suddenly thrown into position 2. The current 'I' through the 10 kΩ resistor as a function of time t, from $t = 0$ is (Give the sketch showing the magnitudes of the current at $t = 0$, $t = t_0$ and $t = \infty$).

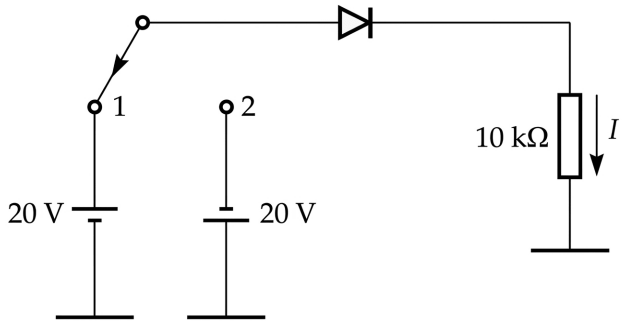


Fig. Q2.91.1

Q2.91.2 MCQ 1991 2M

The small signal capacitance of an abrupt $P^+ n$ junction is 1 nF/cm^2 at zero bias. If the built-in voltage is 1 volt, the capacitance at a reverse bias voltage of 99 volts in nF/cm^2 is

A. 10 **B.** 0.1 **C.** 0.01 **D.** 100

Q2.90.1 MCQ 1990 2M

In a uniformly doped abrupt p-n junction the doping level of the n-side is four (4) times the doping level of the p-side the ratio of the depletion layer width of n-side verses p-side is

A. 0.25 **B.** 0.5 **C.** 1.0 **D.** 2.0

Q2.90.2 MCQ 1990 1M

In a junction diode

A. the depletion capacitance increases with increase in the reverse bias.

B. the depletion capacitance decreases with increase in the reverse bias.

C. the depletion capacitance increases with increase in the forward bias.

D. the depletion capacitance is much higher than the depletion capacitance when it is forward biased.

Q2.89.1 MCQ 1989 2M

The switching speed of P^+N junction (having a heavily doped P region) depends primarily on

A. the mobility of minority carriers in the P^+ region.

B. the lifetime of minority carriers in the P^+ region.

C. the mobility of majority carriers in the N region.

D. the lifetime of majority carriers in the N region.

Q2.89.2 MCQ 1989 2M

In a Zener diode

A. only the P region is heavily doped.

B. only the N region is heavily doped.

C. both P and N regions are heavily doped.

D. both P and P regions are lightly doped.

Q2.88.1 MCQ 1988 2M

In the circuit shown below the current voltage relationship when D_1 and D_2 are identical is given by (Assume Ge diodes)

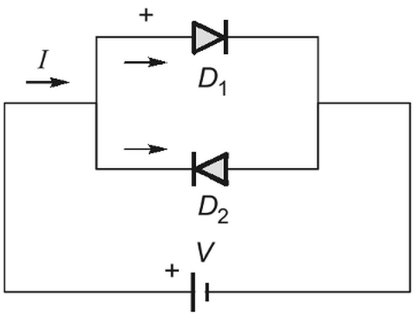


Fig. Q2.88.1

A. $V = \frac{KT}{q} \sinh\left(\frac{I}{2}\right)$

B. $V = \frac{KT}{q} \ln\left(\frac{I}{I_o}\right)$

C. $V = \frac{KT}{q} \sinh^{-1}\left(\frac{I}{2}\right)$

D. $V = \frac{KT}{q} [\exp(-I) - 1]$

Q2.88.2 MCQ 1988 2M

For a pn-junction match the type of breakdown with phenomenon:

List - I:

1. Avalanche breakdown
2. Zener breakdown
3. Punch through

List - II:

A. Collision of carriers with crystal ions

B. Early effect

C. Rupture of covalent bond due to strong electric field.

A. 1-B, 2-A, 3-C **B.** 1-C, 2-A, 3-B
C. 1-A, 2-B, 3-C **D.** 1-A, 2-C, 3-B

Q2.87.1 MCQ 1987 2M

The diffusion capacitance of a *p-n* junction

A. decreases with increasing current and increasing temperature.

B. decreases with decreasing current and increasing temperature.

C. increases with increasing current and increasing temperature.

D. does not depend on current and temperature.

Q2.15.1 C	Q2.15.2 28 to 30	Q2.15.3 2.4 to 2.6
Q2.15.4 A	Q2.14.1 D	Q2.14.2 -32 to -30
Q2.14.3 B	Q2.14.4 A	Q2.13.1 A
Q2.11.1 B	Q2.11.2 D	Q2.10.1 C
Q2.09.1 B	Q2.09.2 B	Q2.08.1 C
Q2.08.2 D	Q2.07.1 A	Q2.07.2 C
Q2.05.1 B	Q2.05.2 B	Q2.04.1 D
Q2.04.2 B	Q2.03.1 C	Q2.02.1 B
Q2.98.1 B	Q2.98.2 B	Q2.95.1 C
Q2.95.2 A	Q2.95.3 D	Q2.93.1 A, B
Q2.91.1 *	Q2.91.2 B	Q2.90.1 A
Q2.90.2 B	Q2.89.1 MTA	Q2.89.2 C
Q2.88.1 B	Q2.88.2 D	Q2.87.1 B

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q2.26.1 C	Q2.26.2 A	Q2.25.1 0.23 to 0.24
Q2.21.1 8.1 to 8.4	Q2.20.1 B	Q2.20.2 MTA
Q2.19.1 34 to 38	Q2.19.2 A	Q2.18.1 0.11 to 0.13
Q2.18.2 0.4 to 0.43	Q2.18.3 D	Q2.17.1 A
Q2.17.2 10	Q2.17.3 D	Q2.16.1 C
Q2.16.2 30 to 34	Q2.16.3 C	Q2.16.4 -5 to -4.6

UNIT

III

BJTs

Topics

1. Introduction to BJTs and Mode of Operations
2. Ideal and Non-Ideal Characteristics of BJTs

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 22	**MCQ:** 14	**MSQ:** 2	**NAT:** 6
1M: 13	**2M:** 9	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MSQ 2026 1M

Figure shows the output characteristics of two different Bipolar Junction Transistors (BJT), BJT 1 with magnitude of Early voltage $|V_{A1}|$, and BJT 2 with magnitude of Early voltage $|V_{A2}|$. Which of the following options is/are correct regarding the Early voltages?

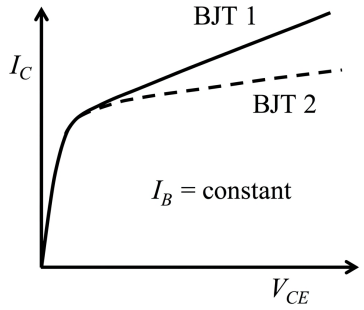


Fig. Q3.26.1

- **A.** $|V_{A1}| > |V_{A2}|$
- **B.** $|V_{A1}|$ is infinitely large
- **C.** $|V_{A1}| < |V_{A2}|$
- **D.** $|V_{A2}|$ is finite

Q3.24.1 MSQ 2024 2M

Which of the following statements is/are true for a BJT with respect to its DC current gain β?

- **A.** Under high-level injection condition in forward active mode, β will decrease with increase in the magnitude of collector current.
- **B.** Under low-level injection condition in forward active mode, where the current at the emitter-base junction is dominated by recombination-generation process, β will decrease with increase in the magnitude of collector current.
- **C.** β will be lower when the BJT is in saturation region compared to when it is in active region.
- **D.** A higher value of β will lead To a lower value of the collector-to-emitter breakdown voltage.

Q3.17.1 MCQ 2017 1M

An npn bipolar junction transistor (BJT) is operating in the active region. If the reverse bias across the base - collector junction is increased, then

- **A.** the effective base width increases and common - emitter current gain increases
- **B.** the effective base width increases and common - emitter current gain decreases
- **C.** the effective base width decreases and common - emitter current gain increases
- **D.** the effective base width decreases and common - emitter current gain decreases

Q3.17.2 MCQ 2017 1M

For a narrow base PNP BJT, the excess minority carrier concentration (Δn_E for emitter, Δp_{E0} for base, Δn_c for collector) normalized to equilibrium minority carrier concentration (n_{E0} for emmitter, p_{B0} for base, n_{C0} for collector) in the quasi-neutral emitter, base and collector regions are shown below. Which one of the following biasing modes is the transistor operating in ?

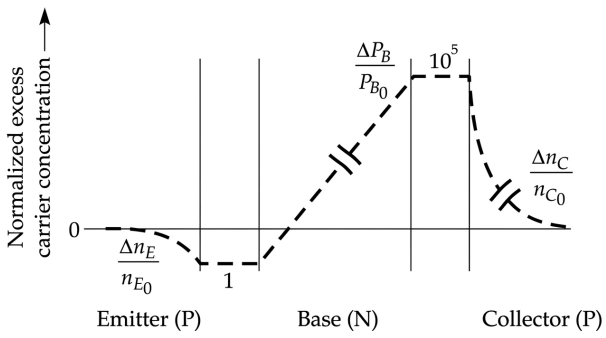


Fig. Q3.17.2

- **A.** Forward active
- **B.** Saturation
- **C.** Inverse active
- **D.** Cutoff

Q3.16.1 NAT 2016 2M

The injected excess electron concentration profile in the base region of an npn BJT, biased in the active region, is linear, as shown in the figure. If the area of the emitter-base junction is 0.001 cm^2, μ_n = 800 $cm^2/(V - s)$ in the base region and depletion layer widths are negligible, then the collector current I_C (in mA) at room temperature is ____ (Given: thermal voltage V_T = 26 mV at room temperature, electronic charge $q = 1.6 \times 10^{-19}$ C)

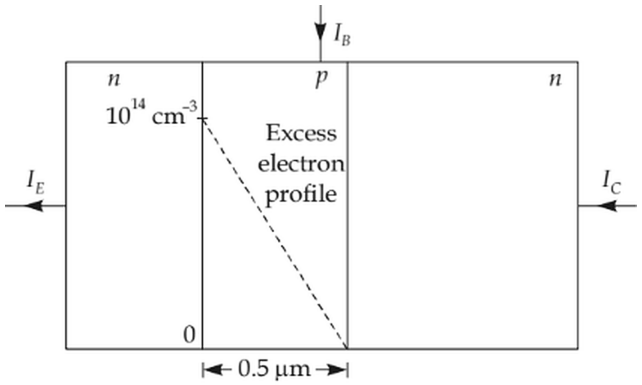


Fig. Q3.16.1

Q3.15.1 NAT 2015 2M

An npn BJT having reverse saturation current $I_S = 10^{-15}$ A is biased in the forward active region with $V_{BE} = 700$ mV. The thermal voltage (V_T) is 25 mV and the current gain (β) may vary from 50 to 150 due to manufacturing variations. The maximum emitter current (in μA) is ____.

Q3.15.2 MCQ 2015 1M

If the base width in a bipolar junction transistor is doubled, which one of the following statements will be TRUE?

A. Current gain will increase

B. Unity gain frequency will increase.

C. Emitter-base junction capacitance will increase

D. Early Voltage will increase

Q3.14.1 NAT 2014 2M

Consider two BJTs biased at the same collector current with area $A_1 = 0.2\,\mu\text{m} \times 0.2\,\mu\text{m}$ and $A_2 = 300\,\mu\text{m} \times 300\,\mu\text{m}$. Assuming that all other device parameters are identical, kT/q = 26 mV, the intrinsic carrier concentration is 1×10^{10} cm^{-3}, and $q = 1.6 \times 10^{-19}$ C, the difference between the base-emitter voltages (in mV) of the two BJTs (i.e., $V_{BE1} - V_{BE2}$) is ____.

Q3.14.2 NAT 2014 2M

A BJT is biased in forward active mode. Assume $V_{BE} = 0.7$ V, kT/q = 25mV and reverse saturation current $I_S = 10^{-13}$ A. The transconductance of the BJT (in A/V) is ____.

Q3.11.1 MCQ 2011 1M

For a BJT the common base current gain $\alpha = 0.98$ and the collector base junction reverse bias saturation current $I_{CO} = 0.6\,\mu$A. This BJT is connected in the common emitter mode and operated in the active region with a base drive current $I_B = 20\,\mu$A. The collector current I_C for this mode of operation is

A. 0.98 mA **B.** 0.99 mA **C.** 1.0 mA **D.** 1.01 mA

Q3.10.1 MCQ 2010 1M

In a uniformly doped BJT, assume that N_E, N_B and N_C are the emitter, base and collector doping in atoms/cm^3, respectively. If the emitter injection efficiency of the BJT is close to unity, which one of the following condition is TRUE

A. $N_E = N_B = N_C$

B. $N_E \gg N_B$ and $N_B > N_C$

C. $N_E = N_B$ and $N_B < N_C$

D. $N_E < N_B < N_C$

Q3.06.1 MCQ 2006 1M

The phenomenon known as "Early Effect" in a bipolar transistor refers to a reduction of the effective base-width caused by

A. Electron - hole recombination at the base

B. The reverse biasing of the base - collector junction

C. The forward biasing of emitter-base junction

D. The early removal of stored base charge during saturation-to-cut off switching

Q3.04.1 MCQ 2004 1M

The neutral base width of a bipolar transistor, biased in the active region, is 0.5 μm. The maximum electron concentration and the diffusion constant in the base are $10^{14}/cm^3$ and $D_n = 25\ cm^2/sec$ respectively. Assuming negligible recombination in the base, the collector current density is (the electron charge is 1.6×10^{-19} - Coulomb)

A. $800A/cm^2$ **B.** $8A/cm^2$
C. $200A/cm^2$ **D.** $2A/cm^2$

Q3.04.2 MCQ 2004 1M

Consider the following statements S1 and S2. S1 : The β of a bipolar transistor reduces if the base width is increased. S2 : The β of a bipolar transistor increases if the dopoing concentration in the base is increased. Which remarks of the following is correct ?

A. S1 is FALSE and S2 is TRUE

B. Both S1 and S2 are TRUE

C. Both S1 and S2 are FALSE

D. S1 is TRUE and S2 is FALSE

Q3.04.3 MCQ 2004 1M

If for a silicon npn transistor, the base-to-emitter voltage (V_{BE}) is 0.7 V and the collector-to-base voltage (V_{CB}) is 0.2 V, then the transistor is operating in the

A. normal active mode **B.** saturation mode
C. inverse active mode **D.** cutoff mode

Q3.99.1 MCQ 1999 1M

The early-effect in a bipolar junction transistor is caused by

A. fast-turn-on

B. fast-turn-off

C. large collector-base reverse bias

D. large emitter-base forward bias

Q3.97.1 MCQ 1997 2M

In a bipolar transistor at room temperature, if the emitter current is doubled, the voltage across its base-emitter junction

A. doubles **B.** halves
C. increases by about 20 mV **D.** decreases by about 20 mV

Q3.96.1 MCQ 1996 2M

If a transistor is operating with both of its junctions forward biased, but with the collector base forward bias greater than the emitter-base forward bias, then it is operating in the

A. forward active mode. **B.** reverse saturation mode.
C. reverse active mode. **D.** forward saturation mode.

Q3.95.1 MCQ 1995 1M

A BJT is said to be operating in the saturation region if

A. both the junctions are reverse biased.

B. base-emitter junction is reverse biased and base-collector junction is forward biased.

C. base-emitter junction is forward biased and base-collector junction is reverse biased.

D. both the junctions are forward biased.

Q3.95.2 MCQ 1995 1M

The breakdown voltage of a transistor with its base open is BV_{CEO} and that with emitter open is BV_{CBO}, then

A. $BV_{CEO} = BV_{CBO}$

B. $BV_{CEO} > BV_{CBO}$

C. $BV_{CEO} < BV_{CBO}$

D. BV_{CEO} is not related to BV_{CBO}

Q3.94.1 NAT 1994 2M

Match the following:

List-I:

A. The current gain of a BJT will be increased.

B. The current gain of a BJT will be reduced.

C. The break-down voltage of a BJT will be reduced.

List-II:

1. The collector doping concentration is increased.
2. The base width is reduced.
3. The emitter doping concentration to base doping concentration ratio is reduced.
4. The base doping concentration is increased keeping the ratio of the emitter doping concentration to base doping concentration constant.
5. The collector doping concentration is reduced.

Q3.92.1 NAT 1992 2M

In a transistor having finite β, forward bias across the base emitter junction is kept constant and the reverse bias across the collector base junction is increased. Neglecting the leakage across the collector base junction and the depletion region generations current, the base current will ____ (increase/decrease/remains constant).

Answer Key

Q3.26.1 C, D	Q3.24.1 A, C, D	Q3.17.1 C
Q3.17.2 C	Q3.16.1 6.55 to 6.75	Q3.15.1 1465 to 1485
Q3.15.2 D	Q3.14.1 378 to 381	Q3.14.2 5.7 to 5.9
Q3.11.1 D	Q3.10.1 B	Q3.06.1 B
Q3.04.1 B	Q3.04.2 D	Q3.04.3 A
Q3.99.1 C	Q3.97.1 C	Q3.96.1 B
Q3.95.1 D	Q3.95.2 C	Q3.94.1 A-2, B-3, C-1
Q3.92.1 decrease		

PrepFusion

UNIT

IV

MOSCap and MOSFETs

Topics

1. Different Modes of Operations of MOSCAPs
2. MOS Physics

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 43	**MCQ:** 29	**MSQ:** 1	**NAT:** 13
1M: 25	**2M:** 18	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2023)–3M(2016)

**See preface for how Avg Weightage and Range are calculated.*

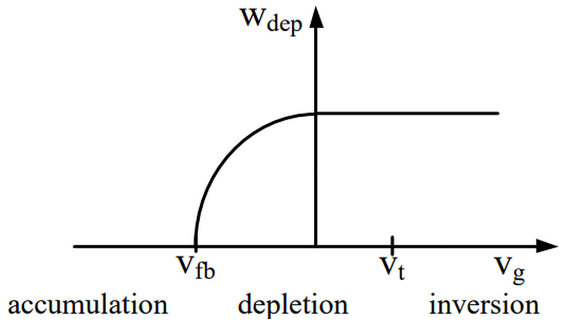
Q4.24.1 NAT 2024 2M

Consider a MOS capacitor made with p-type silicon. It has an oxide thickness of 100 nm, a fixed positive oxide charge of 10^{-8} C/cm^2 at the oxide-silicon interface, and a metal work function of 4.6 eV. Assume that the relative permittivity of the oxide is 4 and the absolute permittivity of free space is 8.85×10^{-14} F/cm. If the flatband voltage is 0 V, the work function of the p-type silicon (in eV, rounded off to two decimal places) is _____.

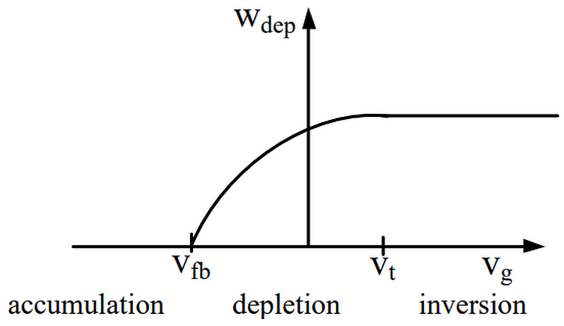
Q4.23.1 MCQ 2023 1M

For a MOS capacitor, V_{fb} and V_t are the flat-band voltage and the threshold voltage, respectively. The variation of the depletion width (W_{dep}) for varying gate voltage (V_g) is best represented by

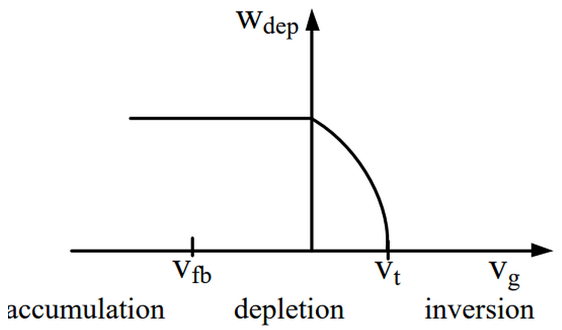
A.



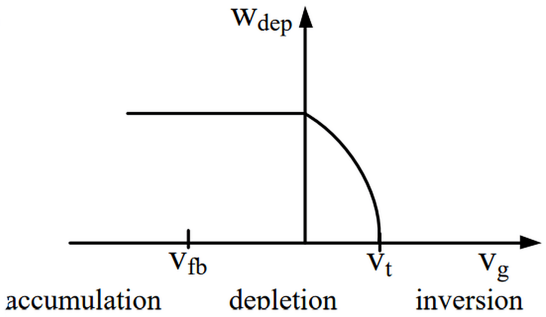
B.



C.



D.



Q4.22.1 NAT 2022 1M

An ideal MOS capacitor (p-type semiconductor) is shown in the figure. The MOS capacitor is under strong inversion with $V_G = 2$ V. The corresponding inversion charge density Q_{IN} is 2.2 μC/cm^2. Assume oxide capacitance per unit area as $C_{OX} = 1.7$ μF/cm^2. For $V_G = 4$ V, the value of Q_{IN} is ____ μC/cm^2 (rounded off to one decimal place).

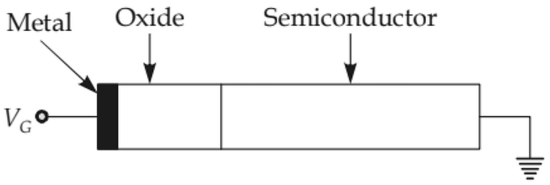


Fig. Q4.22.1

Q4.21.1 MCQ 2021 2M

For an n-channel silicon MOSFET with 10 nm gate oxide thickness, the substrate sensitivity $\left(\frac{\partial V_T}{\partial |V_{BS}|}\right)$ is found to be 50 mV/V at a substrate voltage $|V_{BS}| = 2$ V, where V_T is the threshold voltage of the MOSFET. Assume that, $|V_{BS}| \gg 2\Phi_B$, where $q\Phi_B$ is the separation between the Fermi energy level E_F and the intrinsic level E_i in the bulk.

Parameters given are: Electron charge (q) = 1.6×10^{-19} C , Vacuum permittivity (ε_0) = 8.85×10^{-12} F/m , Relative permittivity of silicon (ε_{Si}) = 12 , Relative permittivity of oxide (ε_{ox}) = 4

The doping concentration of the substrate is

A. 7.37×10^{15} cm^{-3} **B.** 4.37×10^{15} cm^{-3}
C. 2.37×10^{15} cm^{-3} **D.** 9.37×10^{15} cm^{-3}

Q4.20.1 MCQ 2020 2M

The band diagram of a p-type semiconductor with a band-gap of 1 eV is shown. Using this semiconductor, a MOS capacitor having V_{TH} of −0.16 V, C'_{ox} of 100 nF/cm^2 and a metal work function of 3.87 eV is fabricated. There is no charge within the oxide. If the voltage across the capacitor

is V_{TH}, the magnitude of depletion charge per unit area (in C/cm^2) is

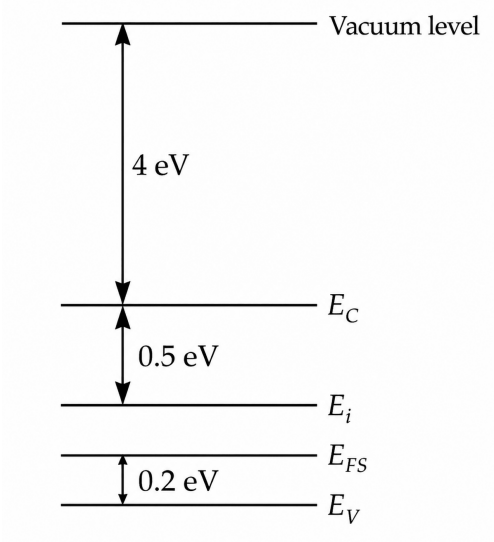


Fig. Q4.20.1

A. 1.7×10^{-8} **B.** 0.52×10^{-8}
C. 1.41×10^{-8} **D.** 0.93×10^{-8}

Q4.19.1 MCQ 2019 1M

The figure shows the high-frequency C-V curve of a MOS capacitor (at T = 300 K) with $\Phi_{ms} = 0$ V and no oxide charges. The flat-band, inversion, and accumulation conditions are represented, respectively, by the points

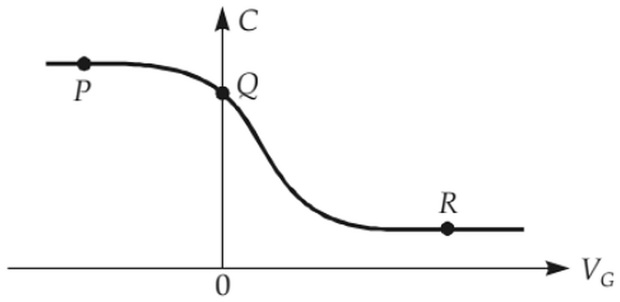


Fig. Q4.19.1

A. P, Q, R **B.** Q, R, P
C. R, P, Q **D.** Q, P, R

Q4.17.1 MCQ 2017 2M

Two n-channel MOSFETs, T1 and T2, are identical in all respects except that the width of T2 is double that of T1. Both the transistors are biased in the saturation region of operation, but the gate overdrive voltage ($V_{GS} - V_{TH}$) of T2 is double that of T1, where V_{GS} and V_{TH} are the gate-to-source voltage and threshold voltage of the transistors, respectively. If the drain current and transconductance of T1 are I_{D1} and g_{m1} respectively, the corresponding values of these two parameters for T2 are

A. $8I_{D1}$ and $2g_{m1}$ **B.** $8I_{D1}$ and $4g_{m1}$
C. $4I_{D1}$ and $4g_{m1}$ **D.** $4I_{D1}$ and $2g_{m1}$

Q4.17.2 NAT 2017 2M

A MOS capacitor is fabricated on p-type Si (silicon) where the metal work function is 4.1 eV and electron affinity of Si is 4.0 eV. $E_C - E_F = 0.9$ eV, where E_C and E_F are the conduction band minimum and the Fermi energy levels of Si, respectively. Oxide $\epsilon_r = 3.9$, $\epsilon_o = 8.85 \times 10^{-14}$ F/cm, oxide thickness $t_{ox} = 0.1\,\mu m$ and electronic charge $q = 1.6 \times 10^{-19}$ C. If the measured flat band voltage of the capacitor is -1V, then the magnitude of the fixed charge at the oxide-semiconductor interface, in nC/cm^2, is ____.

Q4.16.1 NAT 2016 2M

Figures I and II show two MOS capacitors of unit area. The capacitor in Figure I has insulator materials X (of thickness $t_1 = 1$ nm and dielectric constant $\epsilon_1 = 4$) and Y (of thickness $t_2 = 3$ nm and dielectric constant $\epsilon_2 = 20$). The capacitor in Figure II has only insulator material X of thickness t_{Eq}. If the capacitors are of equal capacitance, then the value of t_{Eq} (in nm) is ____

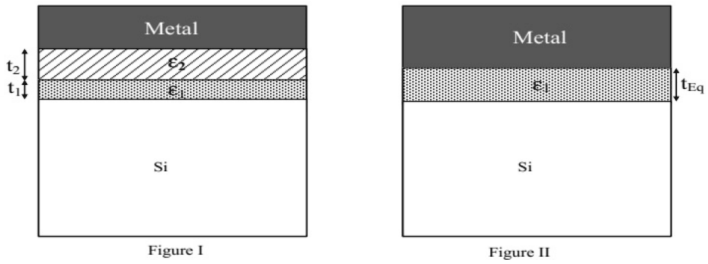


Fig. Q4.16.1

Q4.16.2 MCQ 2016 1M

The figure shows the band diagram of a Metal Oxide Semiconductor (MOS). The surface region of this MOS is in

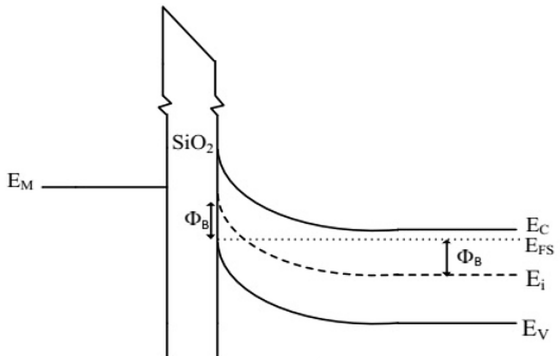


Fig. Q4.16.2

A. inversion
B. accumulation
C. depletion
D. flat band

Q4.16.3 MCQ 2016 2M

A voltage V_G is applied across a MOS capacitor with metal gate and p-type silicon substrate at T=300 K. The inversion carrier density (in number of carriers per unit area) for $V_G = 0.8$ V is 2×10^{11} cm^{-2}. For $V_G = 1.3$ V, the inversion carrier density is 4×10^{11} cm^{-2}. What is the value of the inversion carrier density for $V_G = 1.8$ V?

A. 4.5×10^{11} cm^{-2}
B. 6.0×10^{11} cm^{-2}
C. 7.2×10^{11} cm^{-2}
D. 8.4×10^{11} cm^{-2}

Q4.16.4 MCQ 2016 1M

A long-channel NMOS transistor is biased in the linear region with $V_{DS} = 50$ mV and is used as a resistance. Which one of the following statements is NOT correct?

A. If the device width W is increased, the resistance decreases.

B. If the threshold voltage is reduced, the resistance decreases.

C. If the device length L is increased, the resistance increases.

D. If V_{GS} is increased, the resistance increases.

Q4.16.5 NAT 2016 2M

Consider an n-channel metal oxide semiconductor field effect transistor (MOSFET) with a gate-tosource voltage of 1.8 V. Assume that $\frac{W}{L} = 4$, $\mu_N C_{ox} = 70 \times 10^{-6}$ A V^{-2}, the threshold voltage is 0.3 V, and the channel length modulation parameter is 0.09 V^{-1}. In the saturation region, the drain conductance (in micro seimens) is ____

Q4.16.6 MCQ 2016 1M

Consider the following statements for a metal oxide semiconductor field effect transistor (MOSFET):

- **P**: As channel length reduces, OFF-state current increases.
- **Q**: As channel length reduces, output resistance increases.
- **R**: As channel length reduces, threshold voltage remains constant.
- **S**: As channel length reduces, ON current increases.

Which of the above statements are INCORRECT?

A. P and Q
B. P and S
C. Q and R
D. R and S

Q4.15.1 NAT 2015 2M

The current in an enhancement mode NMOS transistor biased in saturation mode was measured to be 1 mA at a drain-source voltage of 5 V. When the drain-source voltage was increased to 6 V while keeping gate-source voltage same, the drain current increased to 1.02 mA. Assume that drain to source saturation voltage is much smaller than the applied drain-source voltage. The channel length modulation parameter λ (in V^{-1}) is ____.

Q4.15.2 MCQ 2015 2M

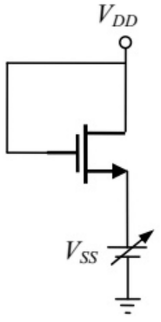
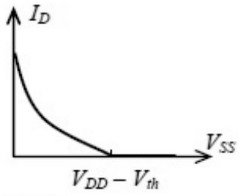
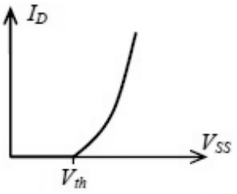
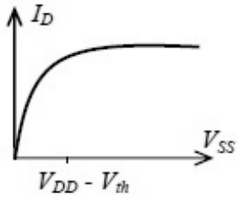


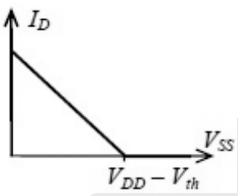
Fig. Q4.15.2

For the NMOSFET in the circuit shown, the threshold voltage is V_{th} , where $V_{th} > 0$. The source voltage V_{SS} is varied from 0 to V_{DD}. Neglecting the channel length modulation, the drain current I_D as a function of V_{SS} is represented by

A.


B.


C.


D.


Q4.15.3 NAT 2015 2M

In a MOS capacitor with an oxide layer thickness of 10 nm, the maximum depletion layer thickness is 100 nm. The permittivities of the semiconductor and the oxide layer are ε_s and ε_{ox} respectively. Assuming $\varepsilon_s/\varepsilon_{ox} = 3$, the ratio of the maximum capacitance to the minimum capacitance of this MOS capacitor is ____.

Q4.15.4 NAT 2015 2M

A MOSFET in saturation has a drain current of 1 mA for $V_{DS} = 0.5$ V. If the channel length modulation coefficient is 0.05 V^{-1}, the output resistance (in $k\Omega$) of the MOSFET is ____.

Q4.14.1 NAT 2014 2M

An ideal MOS capacitor has boron doping-concentration of 10^{15} cm^{-3} in the substrate. When a gate voltage is applied, a depletion region of width 0.5 μm is formed with a surface (channel) potential of 0.2 V. Given $\varepsilon_0 = 8.854 \times 10^{-14}$ F/cm and the relative permittivities of silicon and silicon dioxide are 12 and 4, respectively, the peak electric field (in V/μm) in the oxide region is ____.

Q4.14.2 NAT 2014 2M

The slope of the I_D vs. V_{GS} curve of an n-channel MOSFET in linear regime is $10^{-3}\,\Omega^{-1}$ at $V_{DS} = 0.1$V. For the same device, neglecting channel length modulation, the slope of the $\sqrt{I_D}$ vs. V_{GS} curve (in $\sqrt{\text{A}}$/V) under saturation regime is approximately ____.

Q4.14.3 NAT 2014 2M

A depletion type N-channel MOSFET is biased in its linear region for use as a voltage controlled resistor. Assume threshold voltage $V_{TH} = -0.5$ V, $V_{GS} = 2.0$ V, $V_{DS} = 5$ V, $W/L = 100$, $C_{OX} = 10^{-8}$ F/cm^2 and $\mu_n = 800$ cm^2/V – s. The value of the resistance of the voltage controlled resistor (in Ω) is ____.

Q4.14.4 MCQ 2014 1M

If fixed positive charges are present in the gate oxide of an n-channel enhancement type MOSFET, it will lead to

A. a decrease in the threshold voltage

B. channel length modulation

C. an increase in substrate leakage current

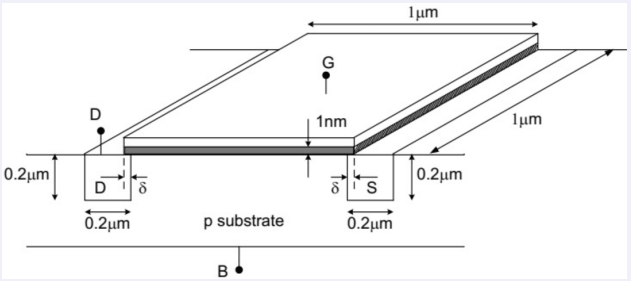
D. an increase in accumulation capacitance

Common Data Questions

The next 2 questions share the data below.

Answer the next two questions using the following data.

In the three dimensional view of a silicon n-channel MOS transistor shown below, $\delta = 20$ nm. The transistor is of width 1 μm. The depletion width formed at every p-n junction is 10 nm. The relative permittivities of Si and SiO_2, respectively, are 11.7 and 3.9, and $\varepsilon_0 = 8.9 \times 10^{-12}$ F/m.



Q4.12.1 MCQ 2012 1M

The source-body junction capacitance is approximately

A. 2 fF **B.** 7 fF **C.** 2 pF **D.** 7 pF

Q4.12.2 MCQ 2012 1M

The gate-source overlap capacitance is approximately

A. 0.7 fF **B.** 0.7 pF **C.** 0.35 fF **D.** 0.24 pF

Q4.12.3 MCQ 2012 1M

The source of a silicon ($n_i = 10^{10}$ per cm^3) n-channel MOS transistor has an area of 1 sq μm and a depth of 1 μm. If the dopant density in the source is $10^{19}/cm^3$, the number of holes in the source region with the above volume is approximately

A. 10^7 **B.** 100 **C.** 10 **D.** 0

Q4.10.1 MCQ 2010 1M

At room temperature, a possible value for the mobility of electrons in the inversion layer of a silicon n-channel MOSFET is

A. 150 $cm^2/V-s$ **B.** 1350 $cm^2/V-s$
C. 1800 $cm^2/V-s$ **D.** 3600 $cm^2/V-s$

Q4.09.1 MCQ 2009 1M

Consider the following two statements about the internal conditions in a n-channel MOSFET operating in the active region.

S1 : The inversion charge decreases from source to drain. S2 : The channel potential increases from source to drain.

Which of the following is correct?

A. Only S2 is true

B. Both S1 and S2 are false

C. Both S1 and S2 are true, but S2 is not a reason for S1

D. Both S1 and S2 are true, and S2 is a reason for S1

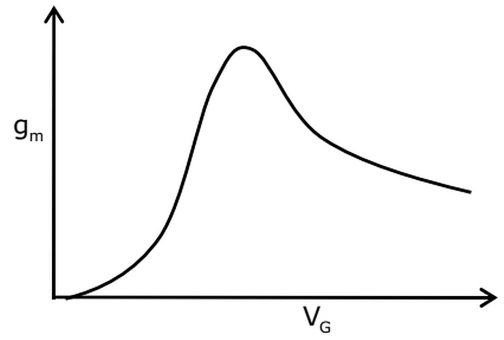
Q4.08.1 MCQ 2008 1M

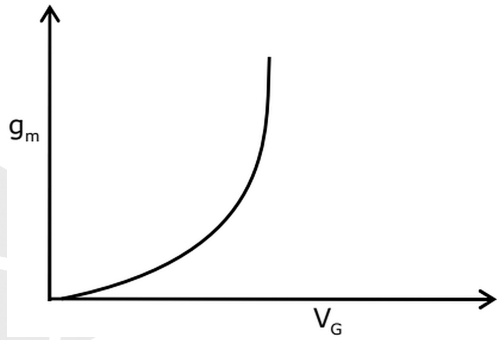
The drain current of MOSFET in saturation is given by $I_D = K(V_{GS} - V_T)^2$ where K is a constant. The magnitude of the transconductance g_m is

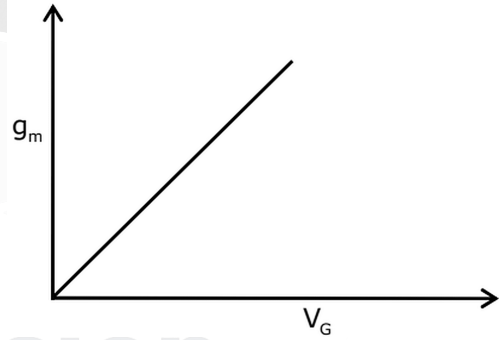
A. $\frac{K(V_{GS}-V_T)^2}{V_{DS}}$ **B.** $2K(V_{GS}-V_T)$
C. $\frac{I_d}{V_{GS}-V_{DS}}$ **D.** $\frac{K(V_{GS}-V_T)^2}{V_{GS}}$

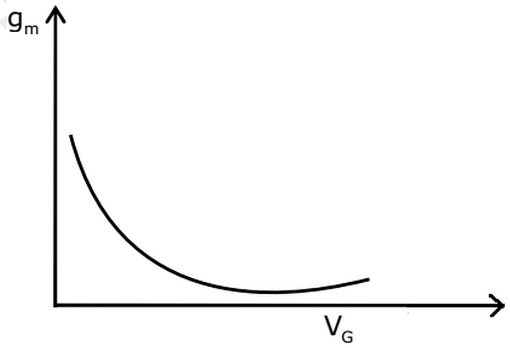
Q4.08.2 MCQ 2008 1M

The measured trans conductance g_m of an NMOS transistor operating in the linear region is plotted against the gate voltage V_G at a constant drain voltage V_D. Which of the following figures represents the expected dependence of g_m on V_G?

A.


B.


C.


D.


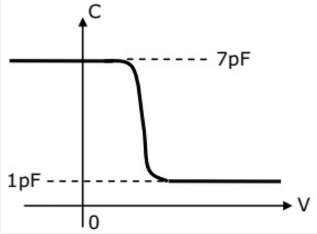
Common Data Questions

The next 3 questions share the data below.

Answer the next three questions using the following data.

The figure shows the high-frequency capacitance-voltage (C-V) characteristics of a Metal/SiO_2/silicon (MOS) capacitor having an area of 1×10^{-4} cm^2. Assume that the permittivities ($\varepsilon_0\varepsilon_r$) of silicon and SiO_2

are 1×10^{-12} F/cm and 3.5×10^{-13} F/cm respectively.



Q4.07.1 MCQ 2007 1M

Consider the following statements about the C-V characteristics plot : S1 : The MOS capacitor has as n-type substrate S2 : If positive charges are introduced in the oxide, the C-V polt will shift to the left. Then which of the following is true?

A. Both S1 and S2 are true

B. S1 is true and S2 is false

C. S1 is false and S2 is true

D. Both S1 and S2 are false

Q4.07.2 MCQ 2007 1M

The maximum depletion layer width in silicon is

A. 0.143 μm **B.** 0.857 μm **C.** 1 μm **D.** 1.143 μm

Q4.07.3 MCQ 2007 1M

The gate oxide thickness in the MOS capacitor is

A. 50 nm **B.** 350 nm **C.** 143 nm **D.** 1 nm

Q4.05.1 MCQ 2005 1M

A MOS capacitor made using p type substrate is in the accumulation mode. The dominant charge in the channel is due to the presence of

A. holes **B.** electrons
C. positively charged ions **D.** negatively charged ions

Q4.04.1 MCQ 2004 1M

Consider the following statements S1 and S2.

S1 : The threshold voltage (V_T) of MOS capacitor decreases with increase in gate oxide thickness.

S2 : The threshold voltage (V_T) of a MOS capacitor decreases with increase in substrate doping concentration. Which Marks of the following is correct ?

A. S1 is FALSE and S2 is TRUE

B. Both S1 and S2 are TRUE

C. S1 is TRUE and S2 is FALSE

D. Both S1 and S2 are FALSE

Q4.03.1 MCQ 2003 1M

For an n - channel enhancement type MOSFET, if the source is connected at a higher potential than that of the bulk (i.e. $V_{SB} > 0$), the threshold voltage V_T of the MOSFET will

A. remain unchanged **B.** decrease
C. change polarity **D.** increase

Q4.01.1 MCQ 2001 1M

The effective channel length of a MOSFET in saturation decreases with increase in

A. gate voltage **B.** drain voltage
C. source voltage **D.** body voltage

Q4.01.2 MCQ 2001 1M

MOSFET can be used as a

A. current controlled capacitor

B. voltage controlled capacitor

C. current controlled inductor

D. voltage controlled inductor

Q4.94.1 NAT 1994 1M

Channel current is reduced on application of a more positive voltage to the GATE of the depletion mode n-channel MOSFET.

(True/False)

Q4.94.2 MCQ 1994 1M

The threshold voltage of an n-channel MOSFET can be increased by

A. increasing the channel carrier concentration.

B. reducing the channel carrier concentration.

C. reducing the GATE oxide thickness.

D. reducing the channel length.

Q4.94.3 NAT 1994 1M

The transit time of a current carriers through the channel of an FET decides its ____ characteristic.

Q4.90.1 MSQ 1990 2M

Which of the following effects can be caused by a rise in the temperature?

A. Increase in MOSFET current (I_{DS})

B. Increase in BJT current (I_C)

C. Decrease in MOSFET current (I_{DS})

D. Decrease in BJT current (I_C)

Q4.89.1 MCQ 1989 2M

In a MOSFET, the polarity of the inversion layer is the same as that of the

A. charge on the gate electrode

B. minority carries in the drain

C. majority carriers in the substrate

D. majority carriers in the source

Q4.88.1 MCQ 1988 2M

In MOSFET devices the N-channel type is better than the P-channel type in the following respects

A. it has better noise immunity.

B. it is faster.

C. it is TTL compatible.

D. it has better drive capability.

Answer Key

Q4.24.1 4.10 to 4.50	Q4.23.1 B	Q4.22.1 5.5 to 5.7
Q4.21.1 A	Q4.20.1 A	Q4.19.1 B
Q4.17.1 B	Q4.17.2 6.85 to 6.95	Q4.16.1 1.55 to 1.65
Q4.16.2 A	Q4.16.3 B	Q4.16.4 D
Q4.16.5 28 to 29	Q4.16.6 C	Q4.15.1 0.018 to 0.026
Q4.15.2 A	Q4.15.3 4.3 to 4.4	Q4.15.4 19 to 21
Q4.14.1 2.3 to 2.5	Q4.14.2 0.06 to 0.08	Q4.14.3 499 to 501
Q4.14.4 A	Q4.12.1 B	Q4.12.2 A
Q4.12.3 D	Q4.10.1 A	Q4.09.1 D
Q4.08.1 B	Q4.08.2 A	Q4.07.1 C
Q4.07.2 B	Q4.07.3 A	Q4.05.1 A
Q4.04.1 D	Q4.03.1 D	Q4.01.1 B
Q4.01.2 B	Q4.94.1 False	Q4.94.2 B
Q4.94.3 switching	Q4.90.1 B , C	Q4.89.1 D
Q4.88.1 B		

UNIT

V

Optoelectronic Devices

Topics

1. Solar Cell
2. Photodiode
3. LEDs

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 12	**MCQ:** 5	**MSQ:** 1	**NAT:** 6
1M: 4	**2M:** 8	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 0M(2016)–4M(2019)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MSQ 2026 2M

Consider an LED based on a direct bandgap semiconductor material with energy bandgap 1.3 eV. Given: Planck's constant, $h = 6.63 \times 10^{-34}$ J s and speed of light in free space is 3×10^8 m s^{-1}. In which of the following wavelength ranges the LED will NOT emit?

- **A.** 1410 ± 20 nm
- **B.** 1090 ± 20 nm
- **C.** 950 ± 20 nm
- **D.** 510 ± 20 nm

Q5.24.1 NAT 2024 2M

The photocurrent of a PN junction diode solar cell is 1 mA. The voltage corresponding to its maximum power point is 0.3 V. If the thermal voltage is 30 mV, the reverse saturation current of the diode (in nA, rounded off to two decimal places) is _____.

Q5.20.1 MCQ 2020 2M

A pn junction solar cell of area 1.0 cm^2, illuminated uniformly with 100 $mW cm^{-2}$, has the following parameters: Efficiency = 15%, open circuit voltage = 0.7V, fill factor = 0.8, and thickness = 200μm. The charge of an electron is $1.6 \times 10^{-19} C$. The average optical generation rate (in $cm^{-3} s^{-1}$) is

A. 0.84×10^{19} **B.** 5.57×10^{19}
C. 1.04×10^{19} **D.** 83.6×10^{19}

Q5.19.1 NAT 2019 2M

A Germanium sample of dimensions 1cmx1cm is illuminated with a 20 mW, 600 nm laser light source as shown in the figure.

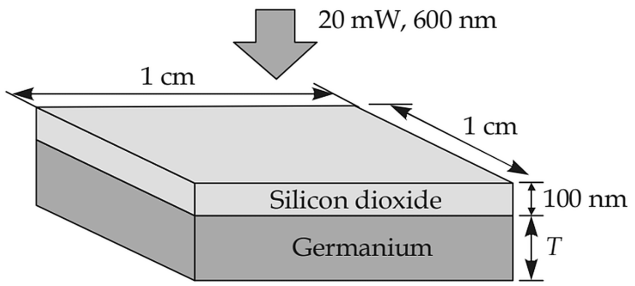


Fig. Q5.19.1

The illuminated sample surface has a 100 nm of lossless Silicon dioxide layer that reflects one-fourth of the incident light. From the remaining light, one-third of the power is reflected from the Silicon dioxide-Germanium interface, one-third is absorbed in the Germanium layer, and one-third is transmitted through the other side of the sample. If the absorption coefficient of Germanium at 600 nm is $3 \times 10^4 cm^{-1}$ and the bandgap is 0.66 eV, the thickness of the Germanium layer, rounded off to 3 decimal places, is ____ μm.

Q5.19.2 MCQ 2019 2M

The quantum efficiency (η) and responsivity (R) at a wavelength λ (in μm) in a p-i-n photodetector are related by

A. $R = \dfrac{\eta\lambda}{1.24}$ **B.** $R = \dfrac{\lambda}{1.24\eta}$
C. $R = \dfrac{1.24\lambda}{\eta}$ **D.** $R = \dfrac{1.24}{\eta\lambda}$

Q5.18.1 NAT 2018 2M

A solar cell of area 1.0cm^2 , operating at 1.0 sun intensity, has a short circuit current of 20mA, and an open circuit voltage of 0.65V. Assuming room temperature operation and thermal equivalent voltage of 26mV, the open circuit voltage (in volts, correct to two decimal places) at 0.2 sun intensity is ____ .

Q5.18.2 MCQ 2018 2M

Red (R), Green (G) and Blue (B) Light Emitting Diodes (LEDs) were fabricated using p-n junctions of three different inorganic semiconductors having different band-gaps. The builtin voltages of red, green and blue diodes are V_R, V_G, V_B respectively. Assume donor and acceptor doping to be the same (N_A and N_D respectively) in the p and n sides of all the three diodes. Which one of the following relationships about the built-in voltages is TRUE?

A. $V_R > V_G > V_B$ **B.** $V_R < V_G < V_B$
C. $V_R = V_G = V_B$ **D.** $V_R > V_G < V_B$

Q5.17.1 NAT 2017 2M

For a particular intensity of incident light on a silicon pn junction solar cell, the photocurrent density J_L is 2.5mA/cm^2 and the open-circuit voltage V_{OC} is 0.451V. Consider thermal voltage V_T to be 25mV. If the intensity of the incident light is increased by 20 times,assuming that the temperature remains unchanged, V_{OC} (in volts) will be ____ .

Q5.16.1 NAT 2016 1M

The figure shows the I-V characteristics of a solar cell illuminated uniformly with solar light of power $100 mW/cm^2$. The solar cell has an area of $3 cm^2$ and a fill factor of 0.7. The maximum efficiency (in %) of the device is ____ .

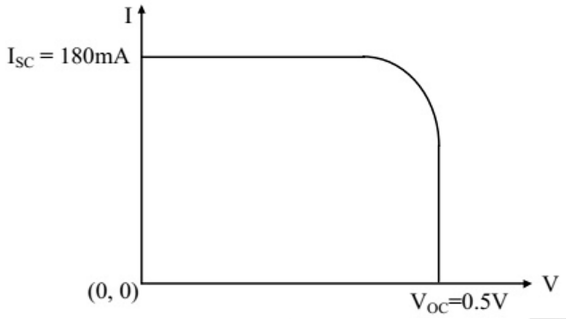


Fig. Q5.16.1

Q5.14.1 NAT 2014 1M

When the optical power incident on a photodiode is 10 μW and the responsivity is 0.8 A/W, the photocurrent generated (in μA) is ____.

Q5.07.1 MCQ 2007 1M

Group I lists three types of p-n junction diodes. Match each device in Group I with one of the option in Group II to indicate the bias condition of the device in its normal mode of operation.

Group I	Group II
(P) Zener Diode	(1) Forward bias
(Q) Solar cell	(2) Reverse bias
(R) LASER diode	
(S) Avalanche Photodiode	

A. P - 1, Q - 2, R - 1 , S - 2

B. P - 2, Q - 1, R - 1 , S - 2

C. P - 2, Q - 2, R - 1 , S - 2

D. P - 2, Q - 1, R - 2 , S - 2

Q5.03.1 MCQ 2003 1M

A particular green LED emits light of wavelength 5490 $\mathring{A}$. The energy bandgap of the semiconductor material used there is (Planck's constant = $6.26 \times 10^{-34}\ Js$).

A. 2.26 eV **B.** 1.98 eV **C.** 1.17 eV **D.** 0.74 eV

Answer Key

Q5.26.1 A, B, D	Q5.24.1 4.00 to 4.26	Q5.20.1 A
Q5.19.1 0.23 to 0.232	Q5.19.2 A	Q5.18.1 0.59 to 0.63
Q5.18.2 B	Q5.17.1 0.51 to 0.62	Q5.16.1 20.5 to 21.5
Q5.14.1 8 to 8	Q5.07.1 B	Q5.03.1 A

Also in this Series

Every PrepFusion GATE title is built the same way: the real previous-year papers, nothing paraphrased or reconstructed, arranged so the pattern the exam repeats is visible on the page. Each questions volume has a companion solutions volume that reuses the same question identifiers, so you can move between the two without a lookup table.

NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

PrepFusion